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Rail Crossing Violation Warning

C-V2X Field Test & Evaluation Report



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Revision History

Revision	Date	Change Description	Affected Sections/Pages
A	09/09/2024	Initial Draft	All
B	1/10/2025	Addition of Regression Test Results	Several throughout the document.

Introduction

There is a need to increase the safety of Highway-Rail Intersections (HRI) beyond traditional active warning devices. A review of Federal Railroad Administration (FRA) safety statistics showed that since 2015¹ there has not been an appreciable improvement in HRI incidents, fatalities, or injuries. Distracted drivers may not notice they are approaching an HRI, fail to perceive activated warning devices, or not recognize that a train is approaching. Adverse atmospheric conditions, engagement with a myriad of distractions (e.g., cell phone use or personal interactions), the influence of alcohol/drugs/medication, or impaired mental capacity due to fatigue or medical events may challenge a vehicle operator's situational awareness.

Beginning in 2015, the Federal Railroad Administration and Battelle effectively produced a Connected Vehicle (CV) system called Rail Crossing Violation Warning (RCVW), designed to notify approaching drivers of an active rail crossing and warn them of a potential collision. The first phase of this project demonstrated a Vehicle-to-Infrastructure (V2I) based technology intended for use at HRIs equipped with active warning devices. When a CV equipped with the RCVW system approached an active HRI (the HRI controller had activated the preemption signal), the system warned the driver via a Driver-Vehicle Interface (DVI) if/when the Vehicle-Based Subsystem (VBS) predicted the driver was not taking sufficient action to prevent a violation. In the third quarter of 2017, Battelle began Phase II of the system, making refinements to the software and hardware, and transforming the proof-of-concept prototype into a more dependable, accurate, and widely applicable system. The updates introduced in Phase II included a refinement of the RCVW warning formula, a functional System Fault Check, utilization of visual and audio alerts based on published human factors design reports, and improved vehicle positional accuracy utilizing Real-Time Kinematic Global Navigation Satellite System Technology.

Phase III (current phase) of the RCVW system replaces the Dedicated Short-Range Communications (DSRC) radios used by both the Roadside-Based Subsystem (RBS) and VBS with Cellular Vehicle-to-Everything (C-V2X) radio technology. On November of 2020², the Federal Communications Commission (FCC) voted and approved to split the 75 megahertz of the DSRC radio spectrum. It allocated the lower 45 megahertz of the band for unlicensed Wi-Fi use and the upper 30 megahertz for the use of transportation related communication technologies. The FCC announced that it will request transportation related operations to transition from DSRC to the emerging C-V2X technology. A specific date of when CV applications should fully phase out DSRC technology and adopt the C-V2X one has not yet been defined. However, in order to be ready by the time the FCC issues a final ruling on the matter, the RCVW architecture has been modified to remove DSRC and incorporate C-V2X technology on both VBS and RBS subsystems. This will allow the system to be ready to transition to the upcoming C-V2X when required.

A Cloud Based Subsystem (CBS) has also been included in the system. This CBS allows the dissemination of the HRI operational and activation status and other attributes to the cloud, providing subscribed authorized users specific HRI and RBS information over the internet.

¹ Source: Federal Railroad Administration Office of Safety Analysis Website (<http://safetydata.fra.dot.gov>).

² Source: Federal Communications Commission, Use of the 5.850-5.925 GHz Band, ET Docket No. 19-308 (<https://docs.fcc.gov/public/attachments/FCC-19-129A2.pdf>)

Testing of the RCVW CBS functionality took place during a bench test activity and the results are presented in the RCVW Cloud-Based Subsystem Bench test report [See Referenced Documents]. The field test activity presented in this document focuses on verifying the overall RCVW capability with a particular focus on the C-V2X related requirements.

The RCVW application, tested by Field Test & Evaluation (FT&E) test cases described herein, provides means for drivers of roadway-vehicles on approach to an HRI to be warned of a predicted violation when the HRI is active.

Document Organization

This document consists of the following chapters and content.

Testing Scope provides an overview and general description of the field test.

Referenced Documents identifies the external documentation referenced within this document.

Summary of RCVW Field Test and Evaluation Approach Results provides an overview of the overall results of the FT&E activities.

Summary of Detected Issues and Anomalies and Potential Refinement describes the issues encountered during FT&E activities and the analysis being conducted to understand it.

Detailed RCVW Field Test & Evaluation Results presents performance results and notes for each test case used to demonstrate the functionality and performance of the RCVW system.

Appendix A. RCVW Requirements Traceability Matrix identifies the test cases wherein each requirement is verified.

Audience for the Document

The intended audiences for this document are developers and those responsible for deploying technology and systems for rail safety and CV systems.

Testing Scope

Document Identification

This document describes the FT&E test cases used during the RCVW field test event. It presents the results and findings as well as the performance of the radio-based system during this testing phase.

The RCVW system went through a detailed Field Test as part of the Phase II of this project back in October of 2020. During Phase-II testing, it was demonstrated that the RCVW system possesses the functionality and performance necessary to deliver the functions and benefits proposed for the system in an operational setting and verified that the RCVW system meets the requirements documented in the System Requirements Specification [2].

The scope of this Phase III field test event was to verify and demonstrate that the C-V2X radios used in the system allow for the RCWV to achieve the necessary performance to deliver the features described in the RCVW Concept of Operations (ConOps) document [1]. Additionally, communication latency requirements were verified to performance of the system was not degraded with the update of the radio communication hardware.

Participation

All functionality tested, as described within this document, was developed by Battelle. Results described within this document were based on testing executed by Battelle at the Battelle's West Jefferson facilities in West Jefferson, OH, on July 25–26, 2024, and on December 12-13, 2025.

Planning

Prior to test execution, Battelle staff installed RCVW equipment, RBS and Roadside Radio Unit (RSU) at a dedicated 0.4-mile test road located in the Battelle's West Jefferson campus. The RCVW RBS (see Roadside-Based Subsystem Hardware) was housed in a traffic cabinet located at the end of the test road (outlined by the red-colored rectangle in Figure 1). An Institute of Electrical and Electronics Engineers (IEEE)-1570 compatible communication system was used to generate the train-detection preemption messages. This device follows the IEEE 1570 standard for the interface between a rail subsystem and a highway subsystem at an HRI. An internet switch was used to connect the radio and RBS over a dedicated internet connection behind Battelle's firewall. This configuration allowed remote connectivity to the RCVW system by the Battelle team which was used to observe system performance and troubleshoot any issues remotely.



Figure 1. Roadside Equipment –(Source: Battelle)

Battelle mounted the C-V2X RSU radio on a metal post located adjacent to the cabinet (outlined by the yellow-colored rectangle in Figure 2). A Real-Time Kinematic (RTK) base station antenna was positioned on a corner on top of the cabinet (outlined by the blue-colored rectangle in Figure 2). Four 110 Vac power outlets within the cabinet were used to provide power to the RBS, RSU (via a Power Over Ethernet (POE) device), switch and IEEE-1570 device.



Figure 2.C-V2X Radio and GNSS Antenna – (Source: Battelle)

A 2009 Mazda 3 vehicle was used for all the field test cases. See Figure 3.

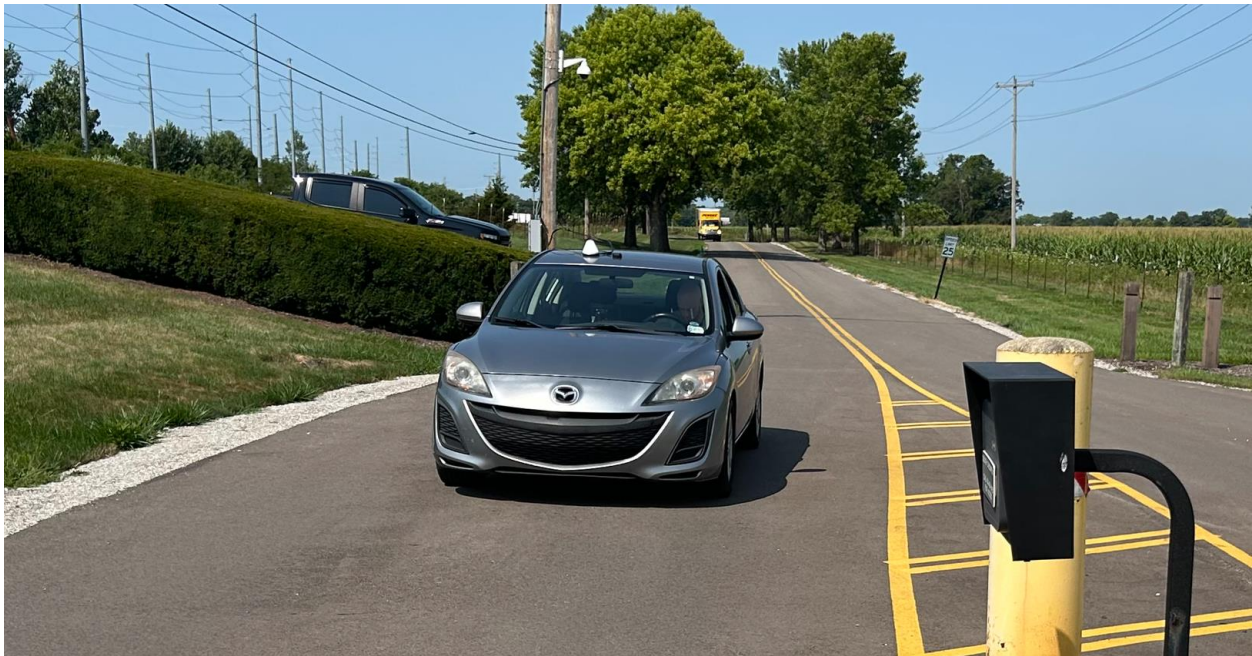


Figure 3. RCVW/Test Vehicle – (Source: Battelle)

Figure 4 shows the RCVW's DVI (outlined by the red-colored rectangle). The rest of the RCVW VBS components were located in the back of the car See Figure 5.



Figure 4. RCVW Equipment on Vehicle – (Source: Battelle)



Figure 5. RCW VBS Plate – (Source: Battelle)

Figure 6 shows the RCW's antennas as placed on the roof of the vehicle. The C-V2X radio and Global Navigation Satellite System (GNSS) antennas are contained within the unit shown in the blue box.



Figure 6. RCVW/Test Equipment on Roof of Vehicle – (Source: Battelle)

To allow for remote connectivity to the VBS, a Verizon MiFi 8800L 4 g Jetpack internet hotspot. was used.

The test was conducted by three Battelle team members. One Battelle team member monitored the algorithm operation, remotely accessing the Vehicle's RCVW systems when needed for any technical aspects and resolution of issues, and assisted when features needed to be turned off. A second Battelle team member was responsible for collecting the start and end times for each test run start and ending, notifying the CV driver of the test to be performed, taking notes of any

irregularities communicated by the driver and support in reconciliation activities. A third team member was dedicated to operating the vehicle, which involved driving the vehicle straight along the approach, reaching and maintaining the designated vehicle speed. A speed of 35 mph was selected after several test runs were conducted and the driver observed that higher speeds could pose a safety concern for vehicle control and deceleration (due to the test track geometry).

Prior to the field test efforts, Battelle staff marked several locations along the test track of particular interest:

- The locations of the stop bar.
- Vehicle locations where the HRI activation should be manually turned on or off as the vehicle reached them. (test cases 1.3,1.4,1.5 and1.6).

At the conclusion of each day of testing, the Battelle team retrieved system log files to analyze the latency and positional accuracy of the system (see C-V2X Messaging and Data Processing Latency Requirement Verification). In addition, detailed notes from each day capturing unexpected or abnormal test results were used to generate a more focused approach for subsequent testing.

RCVW FT&E Test Setup

The following sections describe the layout of equipment and connections used for the RCVW VBS and RBS components.

The RCVW system consists of two physically separate subsystems: a VBS installed in the CVs and an RBS integrated with roadside infrastructure at HRIs. Both subsystems share some common hardware and software components, as well as include unique components. The RCVW system in its entirety includes the following components:

- **Computing Platform (CP):** The heart of the RCVW system are its CPs. The CPs control the RCVW subsystems. One is located in the VBS and a second in the RBS. The RCVW software resides within both CPs.
- **C-V2X Radios:** The VBS and RBS utilize C-V2X radios, an On-Board Unit (OBU) and an RSU, to provide low-latency wireless communications.
- **GNSS:** A multi-band GNSS module with built-in RTK technology resides within the VBS to provide real-time lane-level position data. A similar device resides within the RBS to provide Radio Technical Commission for Maritime Services (RTCM) corrections to be broadcasted via C-V2X.
- **The DVI** is the RCVW interface to the connected vehicle driver which provides the RCVW warnings and inform alerts.

Hardware Overview

Roadside-Based Subsystem Hardware

A hardware block diagram for the RCVW RBS used during field test is shown in Figure 7.

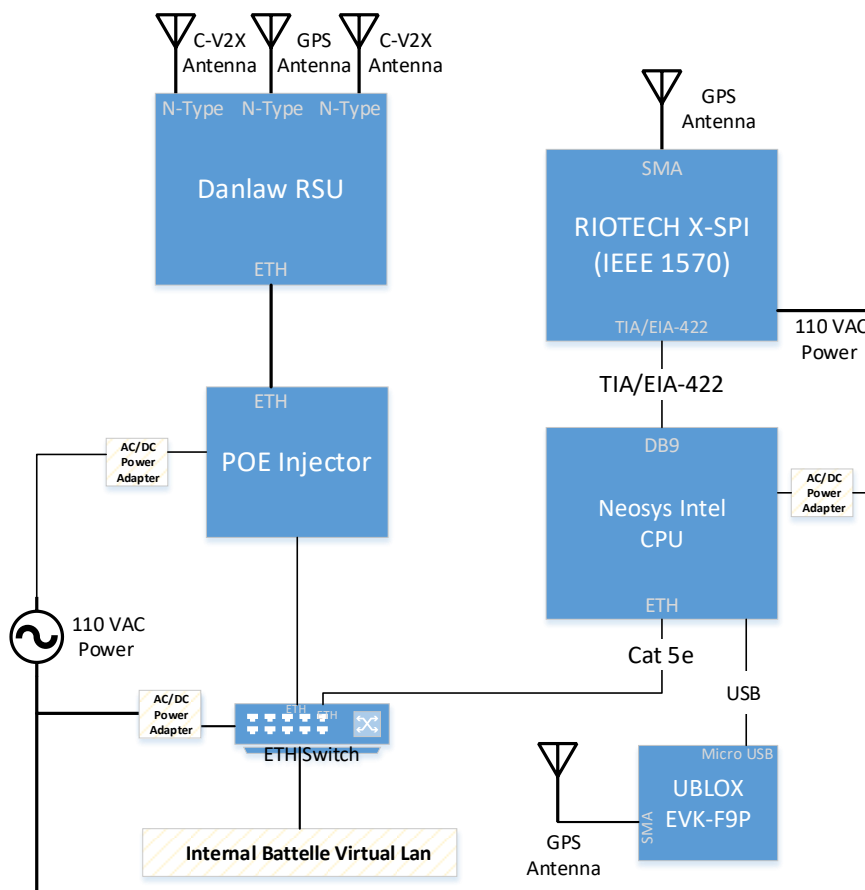


Figure 7. RBS Hardware Used for Field Test- (Source: Battelle)

The RCVW RBS produced SAE J2735 MAP Data (MAP) and Signal Phase and Timing (SPaT) messages as well as RTCM messages to support the RCVW VBS algorithms used to provide alerts to the driver³. The J2735 MAP file contained the Approach Zone geometry for the mock-equipped HRI, while the J2735 SPaT message communicated the HRI activity status (i.e., whether the warning devices were active). The RTCM messages broadcasted contain real-time details about the GNSS network, as well as perturbations in the ionosphere and troposphere

³ Issues with the application of the RTCM corrections were experienced during the test event. See Summary of Detected Issues and Anomalies and Potential Refinement section.

which are to be used by the vehicles onboard GNSS RTK unit to correct the location solution of the vehicle.

RBS Equipment:

- C-V2X RSU
 - A Danlaw RouteLink RSU
- POE injector to supply the RSU with power:
 - Transition Networks SI-ES-1200-LRT
 - Mean Well MDR-60-48 Power Supply
- Central Processing Unit (CPU) running the V2X Hub, MAP, HRI Status, Command, and C-V2X Message Receiver plugins
 - A Neousys POC-451 VTC with Intel Atom x6425E quad-core processor
- GNSS base station providing RTCM corrections
 - U-blox EVK-F9P application board
- IEEE 1570 preemption device
 - RIOTECH X-SPI
- Ethernet Switch
 - Linksys SE3005: 5-Port Gigabit Ethernet Unmanaged Switch

RBS Connections:

- The Neousys Intel CPU, ethernet switch, PoE injector and IEEE 1570 Serial interface received power from the power outlet located within the signal controller cabinet.
- C-V2X antennas were connected to the RSU
- GNSS antennas were connected to the RSU, the RIOTECH X-SPI device and the EVK-F9P GNSS device
- U-blox EVK-F9P application board RTK GNSS module was connected to the Neousys Intel CPU via a vendor provided USB cable
- Ethernet cables were run and landed as follows:
 - Neousys CPU to the ethernet switch
 - PoE injector data /IN port to ethernet switch
 - PoE injector P+D / OUT port to the RSU
- The IEEE 1570 Serial communication interface connected to the CPU via a TIA/EIA-422 compliant cable.

Vehicle-Based Subsystem Hardware

A hardware block diagram for the RCVW VBS used during field test is shown in Figure 8.

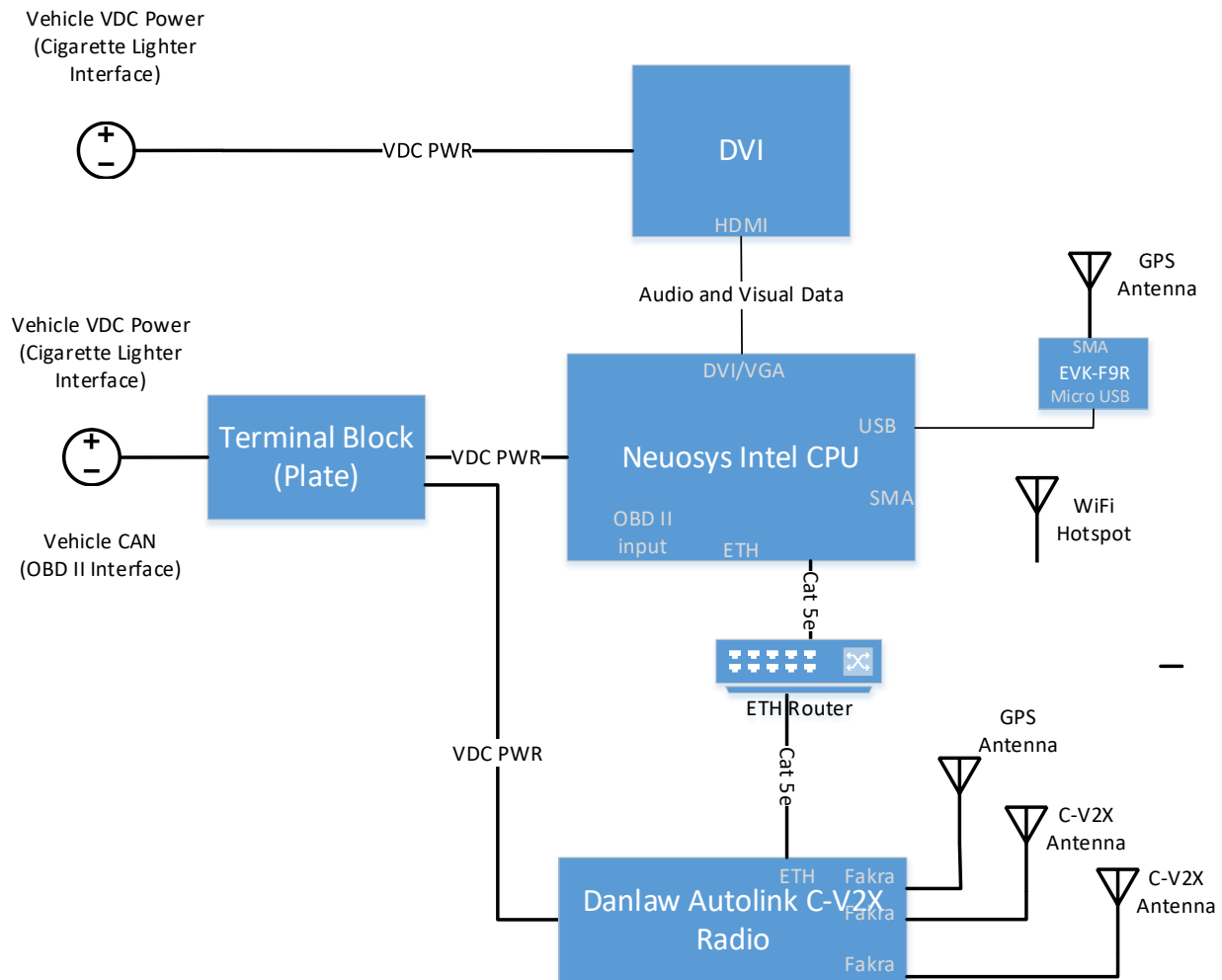


Figure 8. VBS Hardware Used for Field Test - (Source: Battelle)

VBS Equipment:

- CPU running the V2X Hub, RCVW, Differential GNSS, C-V2X Message Receiver plugins
 - Neosys Nuvo-7204VTC Supporting Intel® 9th/ 8th-Gen CPU
- CV2-X OBU
 - Danlaw Autolink OBU
 - External magnet mount antenna for C-V2X
- GNSS
 - U-blox EVK-F9R application board
 - External magnet mount antenna for GNSS

- Wi-Fi Hotspot
 - MiFi 8800L 4 g Jetpack used to enable remote communications to the VBS system
- DVI
 - TRU-Vu SRMOBTC-8.4R-BAT digital monitor
 - RAM Pod HD vehicle mount to support the DVI
 - DVI Power Cable

VBS Connections:

- The DVI was attached to the RAM Pod HD vehicle mount hardware. This hardware is designed to be bolted to the vehicle's passenger seat rail
- The DVI receives power from a cigarette lighter interface
- The DVI connects to the Neousys CPU via a High-Definition Multimedia Interface (HDMI)-to DVI cable
- The Neousys CPU, Danlaw, Autolink OBU and GNSS device receive power from a cigarette lighter interface
- Antennas are attached to the roof of the vehicle and connected to the Danlaw autolink and GNSS devices
- ZED-EVK-F9R RTK GNSS module was connected to the Neousys CPU intel CPU via a vendor provided USB cable
- Danlaw Autolink connects to the CPU via an Ethernet cable

Referenced Documents

Referenced Documentation

1. Sanchez-Badillo, A., Baumgardner, G (2023). Rail Crossing Violation Warning - Concept of Operations. Federal Railroad Administration,
2. Sanchez-Badillo, A., Baumgardner, G (2023). Rail Crossing Violation Warning. System Requirements Specification. Federal Railroad Administration.
3. Baumgardner, G., Sanchez-Badillo, A. (2020). Rail Crossing Violation Warning - Architecture and Design Specifications.
4. American Association of State and Highway Transportation Officials. (2018). A Policy on Geometric Design of Highways and Streets. AASHTO..

Summary of RCVW Field Test and Evaluation Approach Results

The objective of the verification testing event was to verify and demonstrate that the RCVW system possesses the functionality and performance necessary to deliver the functions and benefits proposed for the system described in the RCVW Concept of Operations [1]. The intent of FT&E testing is to verify that the RCVW system meets the requirements documented in the System Requirements Specification [2] via controlled field testing of the design specified in the Architecture and Design Specifications document [3].

This chapter begins with a summary of the approach taken in conducting the testing, provides test execution and coordination details, presents a summary of testing performance and concludes with a summary of issues found and resolutions implemented during FT&E testing. The next section presents detailed test summary results and notes.

Approach to Scenario-Based Field Testing

The RCVW system has been upgraded by replacing its DSRC radios with C-V2X communications radios.

The scope of FT&E was to assess the performance of the RCVW system after the radio update. The RCVW system was fully tested during Phase-II of this project, ⁴ the scope of the current Field Test effort covers only a subset of the scenario-based test cases performed in RCVW Phase 2 which are summarized on Table 2. These tests were performed in a test track where the RCVW system was integrated into surrogate infrastructure. All scenario-based tests used a real vehicle driven towards a mock equipped HRI, configured as described above.

A total of 14 scenario-based test cases were performed; 11 consisted of 3 runs approaching at 35 mph, 1 consisted of the vehicle travelling outside of the test track to measure radio signal coverage, 1 test case had the vehicle approach, stop and remain within the HRI hazard zone and 1 stability test was performed in three stages which were aimed at assessing the level of nuisance alerting observed with the vehicle stopped at various locations near/within/beyond the HRI Hazard Zone. Of the 14 test cases, 5 were fault verification test cases.

The following parameters applied across all scenario-based test cases:

- Only a sedan type vehicle was used for the totality of test cases
- The vehicle was a Type A (passenger vehicle) 2009 Mazda 3
- The approach lane was 1,275 ft (388.62 m).
- The CV approach speed was 35 mph.
- All test cases were on Dry pavement conditions.
- All scenario-based test cases involved driving the vehicle straight along the approach except the following

⁴ For a full report of RCVW Phase-II see: <https://railroads.dot.gov/elibrary/rail-crossing-violation-warning-rcvw-application-phase-ii>

- Test case 2.1 involved the vehicle stopped within the HRI Hazard zone. While the vehicle was static within the hazard zone, the preemption signal was toggled to observe the different alerts when preemption is active vs deactivated.
- Test case 4.1 specified having the CV approaching the HRI location from adjacent roads to quantify the comms range of the radio signal.
- Test case 4.2 required the CV to test the stability of the GNSS on board device by placing the vehicle on separate locations within/outside the HRI Hazard zone and observing any changes in the inform alerts/warning alerts.
- All scenario-based test cases began with the HRI in the active state, except the following
 - Test Case 1.1, the CV approaches and travels through the HRI with the HRI not active during the entire test case.
 - Test Case 1.3, 1.4 and 1.5 began with the HRI in the inactive state and as the vehicle reaches the $\frac{3}{4}$ approach zone point, midpoint, and near the HRI hazard zone the HRI becomes active.
 - Test Case 4.1 and 4.2 does not require the HRI to be active.
- Test Case 1.6 began with the HRI in the active state and as the vehicle reaches the midpoint, HRI becomes inactive.
- Three traffic cones were used along the Approach Zone only for visual reference while conducting tests.
 - Cone #1: $\frac{3}{4}$ approach zone
 - Cone #2 midpoint zone
 - Cone # 3: Near the HRI
- One track location was used during field test activities (see Figure 9)
 - Latitude: 39.956447
 - Longitude: -83.250639

Note: Icons in Figure 9 are for illustration purposes and do not reflect the actual locations of the cones.

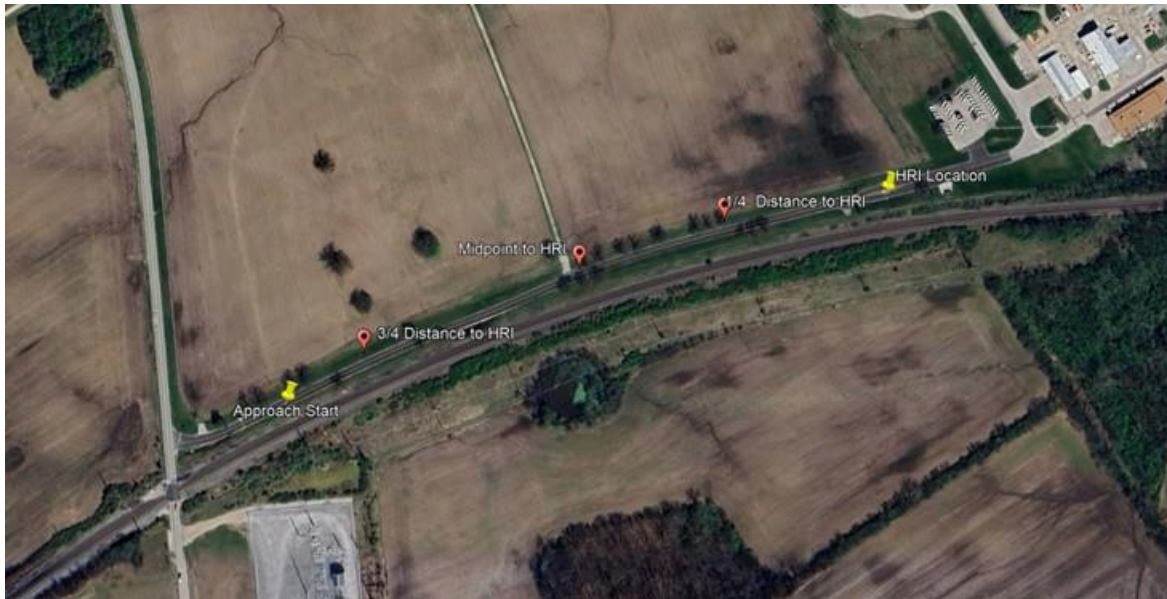


Figure 9. Flat Course - (Source: Google Maps)

- All tests were performed using an IEEE 1570-compliant serial interface to simulate grade crossing warning devices activation status.
- Voltage based grade crossing status reception was not performed during this round of testing.
- The American Association of State Highway and Transportation (AASHTO) Green Book [4] recommends the use of 2.5 seconds perception reaction time. This value reflects observed behavior for the 90th percentile driver
- The following system parameters were used during testing. These are all configurable parameters and can be modified according to the system needs. Once RCVW starts default values are assigned. See Table 1.

Table 1. RCVW System Parameters

Key	Value	Description
Message Expiration	2000	Parameter used to enter the amount of time in milliseconds to wait before issuing a warning alert indicating that current message data is stale.
Output Interface	0	Parameter used to input the value that corresponds to the type of interface that the application needs to display its messages on. 0=Digital Visual Interface (DVI), 1=Ford SYNC, 2=Android Auto.
Distance to HRI	480	Parameter that indicates the maximum distance (in meters) away from an equipped HRI that a System Fault may be issued due to communication failure. When the VBS is within this distance of the HRI, the system checks for MAP and SPAT expiration timeout, if the GNSS system achieved an RTK fix and if location messages are being received at the configured rate.

Key	Value	Description
HRI Locations	["Latitude": 39.9570244, "Longitude": -83.2478429, "HRIName": "Battelle West Jefferson"]	This parameter shows the Latitude, Longitude, and name of the HRI loaded into the system during field testing.
Extended Intersection	0.1	The percentage to add to the radius of the intersection divided by 100. i.e., in this case the percentage to be added is 10%. So, the value to enter is $10/100 = 0.1$
HRI Warning Threshold Speed	1.0	The maximum vehicle speed in meters per second for which the HRI warning alert will be active if the vehicle is in the HRI and moving. If the vehicle's speed falls below this threshold, a warning alert will be issued to the driver.
Use Calculated Deceleration	false	Use calculated deceleration to determine if vehicle will stop before HRI in addition to velocity-based warning calculation.
Log Level	DEBUG	The logging level of the RCVW system.
V2 Antenna Placement X	0.8128	Antenna placement X with respect to front left corner of vehicle in meters.
V2 Antenna Placement Y	2.286	Antenna placement Y with respect to front left corner of vehicle in meters.
V2 Antenna Placement Height	1.651	Antenna height with respect to the road surface in meters.
V2 GPS Error	3.12	Global Positional System (GPS) longitudinal error in meters. The system uses this value directly in the RCVW calculation formula to calculate the issuing of inform and warning alerts. This value represents the longitudinal error of the GNSS system. It is a configurable parameter and as such it can be modified according to the system and vehicle performance. The default value of 3.12 was selected as a result of preliminary testing to show good system performance.
V2 Reaction Time	2.5	Perception-Reaction time in seconds. AASHTO uses the term "Perception-reaction" time and it represents the time it takes for a road user to 1) realize that action is needed due to a road condition, 2) decide what action to take and 3) start the action.
V2 Communication Latency	0.3	Communication latency in seconds. The system uses this value directly in the RCVW calculation formula to calculate the issuing of inform and warning alerts. This parameter accounts for C-V2X radio signal communication latencies and IEEE 1570 data package reception latencies (if used). The default value of 0.3 is based upon RCVW system requirements VBS-022 and RBS-021 (see the Appendix. RCVW Requirements Traceability Matrix) for allowable communication latency.
V2 Application Latency	0.085	Application latency in seconds. The system uses this value directly in the RCVW calculation formula to

Key	Value	Description
		calculate the issuing of inform and warning alerts. It considers the latency of the whole RCVW application for processing data and issuing inform and warning alerts. The default value of 0.085 is based upon RCVW system requirement VBS.0-23(see the Appendix. RCVW Requirements Traceability Matrix for allowable application latency.
V2 Deceleration Car	3.4	Minimum expected controlled deceleration for a car in $\frac{m}{s^2}$.
V2 Deceleration Light Truck	2.148	Minimum expected controlled deceleration for a light truck in in $\frac{m}{s^2}$
V2 Deceleration Heavy Truck	2.322	Minimum expected controlled deceleration for a heavy truck in in $\frac{m}{s^2}$
V2 Vehicle Type	1	Vehicle type, 1 = Car, 2 = Light Truck, 3 = Heavy Truck.
V2 Vehicle Length	4.8	The length of the vehicle in meters.
V2 Use Vehicle Based Measurement (VBM) Deceleration	false	Use VBM deceleration to determine if vehicle will stop before HRI in addition to velocity-based warning calculation.
V2 Log SPaT	false	Log SPaT messages at DEBUG level.
V2 Critical Message Expiration	500	The amount of time in milliseconds to wait before issuing a warning that the current critical message data is stale.
V2 Use Config Grade	false	If False, the system will use the grade directly from the receiving MAP. If True, the system will use the V2 Grade configurable variable for grade calculations.
V2 Grade	0	If Parameter V2 Use Config Grade is set to True, this grade value will be used in warning distance calculations. The value is defined as change in height over change in distance.
V2 Check RTK	false	If enabled check location message for RTK fix while in range of HRI. If enabled and the location message does not show an RTK fix, a System Fault is issued.
V2 Check Location Frequency	true	If set to true, the system will check the Location Message reception rate. If the rate of location messages falls below the value of the parameter shown in V2 Minimum Location Frequency, a System Fault message is issued.
V2 Location Frequency Sample Size	30	This parameter is used if V2 Check Location Frequency is set to true. It is the number of location messages to sample to determine frequency.
V2 Minimum Location Frequency	4	This is the minimum allowed average location message frequency in messages per second.

Key	Value	Description
V2 Max Heading Change	45	The maximum allowed heading change in degrees before ignoring the new position.
V2 Max Ignored Positions	7	The maximum number of consecutively ignored positions due to heading change.

Source: Battelle

RCVW uses the current AASHTO Green Book Stopping Sight Distance formula to issue a warning alert to a vehicle, modified to include additional information unique to the current implementation.

$$D = N + P + 0.278 * V * (T[r] + T[c] + T[a]) + (0.039 * V^2)/(A/g+G) \quad (1)$$

here:

D = distance (m)

N= antenna placement (m)

P = GPS error (m)

V = speed (m/s)

T[r] = reaction time (s), AASHTO 90% rate = 2.5 s

T[c] = communication latency (s)

T[a] = application latency (s)

A = acceleration (m/s²), AASHTO 90% rate = $3.4 \frac{m}{s^2}$

G = grade, rise/run (m/m)

g = $9.81 \frac{m}{s^2}$

For this FT&E task, the Battelle team decided to configure the following latency parameters:

- T[c] = 0.3 seconds
- T[a] = 0.085
- P = 3.12 m

These values were chosen based on the maximum allowed latency according to the RCWV system requirements and were assigned to observe the response of the system. For the case of the GPS error, the value of 3.12 m represents a longitudinal error, it was selected as a result of testing performed in past projects to show good system performance.

The standard perception-reaction time of 2.5 seconds cover 90 percent of all situations, however the green book also covers situations that are more complex or critical and has guidelines for increasing this time to anywhere from 3 seconds to 9.1 seconds for different environments. A comfortable deceleration rate of 3.4 m/s² also covers 90 percent of all drivers in both dry and wet conditions given current pavement, tire, and vehicle technologies. The deceleration rates used are based on the current Federal Motor Vehicle Safety Standards (FMVSS) 121 and FMVSS 135.

A ratio between the AASHTO guidelines for stopping sight distance and the FMVSS requirement for passenger cars in ideal conditions was calculated and the same ratio was used to generate recommended deceleration values to use.

Table 2 summarizes the conditions and inputs for each of the scenario-based test cases executed in FT&E.

Table 2. Distinctive Conditions for the Scenario-Based FT&E Test Cases

Test Case Number	CV Location at Preemption Signal Detection	CV Speed at Approach Entry	RCVW State
1.1	N/A No preemption	35 mph	Normal
1.2	Start of Approach	35 mph	Normal
1.3	¾ point	35 mph	Normal
1.4	Midpoint	35 mph	Normal
1.5	Near HRI Hazard Zone	35 mph	Normal
1.6	Start of Approach, then preemption is removed at Midpoint	35 mph	Normal
2.1	Stopped on Tracks (preemption will deactivate during the test)	N/A	Normal
3.1	Start of Approach	35 mph	Fault – C-V2X
3.2	N/A	35 mph	Fault – C-V2X
3.3	Start of Approach	35 mph	Fault – GPS
3.4	Start of Approach	35 mph	Fault – MAP
3.5	Start of Approach	35 mph	Fault – IEEE-1570
4.1	N/A No Preemption required	N/A	Normal
4.2	N/A No Preemption Required	N/A	Normal

Source: Battelle

Scenario-Based Test Execution and Coordination

RBS integration at the mock railroad crossing along with vehicle configuration took place on July 25 and 26 2024. A series of trial runs were performed this day to 1) understand vehicle and system performance, 2) driver performance and 3) to practice overall duties for all team members. These trial runs additionally allowed the driver to understand the different inform and warning alerts. the system would issue during testing.

Table 3 below presents a chronologically sequenced characterization summary of the FT&E scenario-based testing activities.

Table 3. Chronological Summary of Test Performed During FT&E

Date	Test Case #	Initial Vehicle Speed	HRI Active Initiation	RCVW State	Note
7/25/24	1.1	35 mph	N/A No preemption	Normal	Run presented no errors.
	1.2	35 mph	Start of Approach	Normal	Run presented no errors.
	2.1	N/A	Start of Approach	Normal	The system did not issue a different message with the vehicle in the HRI and preemption ON/OFF
7/26/24	1.3	35 mph	$\frac{3}{4}$ point	Normal	Run presented no errors.
	1.4	35 mph	Midpoint	Normal	Run presented no errors.
	1.5	35 mph	Near HRI Hazard Zone	Normal	Run presented no errors.
	1.6	35 mph	Start of Approach, then preemption is removed at Midpoint	Normal	Run presented no errors.
	3.1	35 mph	Start of Approach	Fault – C-V2X	After the C-V2X RBS radio was turned off, the DVI displayed flashing fault messages rather than a stable one. The software clears and re-issues the fault messages This issue was known prior to test, run presented no unexpected errors.
	3.2	35 mph	Start of Approach	Fault – C-V2X	After entering MAP, the DVI displayed flashing fault messages rather than a stable one. The software clears and re-issues the fault messages This issue was known prior to test, run presented no unexpected errors
	3.4	35 mph	Start of Approach	Fault – MAP	After MAP broadcast was turned off, the DVI displayed flashing fault messages rather than a stable one. The software clears and re-issues the fault messages This issue was known prior to test, run presented no unexpected errors

Date	Test Case #	Initial Vehicle Speed	HRI Active Initiation	RCVW State	Note
	3.5	35 mph	Start of Approach	Fault – IEEE-1570	After IEEE-1570 was disconnected the DVI displayed flashing fault messages rather than a stable one. The software clears and re-issues the fault messages. This issue was known prior to test, run presented no unexpected errors.
7/26/24	4.1	N/A	N/A No Preemption required	Normal	The System was configured with the incorrect distance to HRI radius parameter. This resulted in not being able to see any radio range related errors in the DVI. However, radio ranges were calculated using the system log files.
	4.2	N/A	N/A No Preemption required	Normal	Run presented no errors.
	3.3	35 mph	Start of Approach	Fault – GPS	RTK messaging was unavailable for all testing, this issue was known beforehand. To conduct testing, the RTK Fault check was disabled. This check was enabled for Test case 3.3. No issues were encountered.
12/12/2024	1.2	35 mph	Start of Approach	Normal	Run presented no errors.
	3.1	35 mph	Start of Approach	Fault – C-V2X	The DVI displayed flashing fault messages rather than a stable one. The software clears and re-issues the fault messages. The proper fault was issued.
	3.3	35 mph	Start of Approach	Fault – GPS	The DVI displayed flashing fault messages rather than a stable one. The software clears and re-issues the fault messages. Issues were seen throughout the test event where the system had issues to hold an RTK fix. The proper fault was issued.
	3.4	35 mph	Start of Approach	Fault – MAP	The DVI displayed flashing fault messages rather than a stable one. The software clears and re-issues the fault messages. The proper Fault was issued.
12/13/2024	1.3	35 mph	$\frac{3}{4}$ point	Normal	Run presented no errors.
	2.1	N/A	Start of Approach	Normal	The system did not issue a different message with the vehicle in the HRI and preemption ON/OFF

Date	Test Case #	Initial Vehicle Speed	HRI Active Initiation	RCVW State	Note
	3.5	35 mph	Start of Approach	Fault – IEEE-1570	After IEEE-1570 was disconnected the DVI displayed flashing fault messages rather than a stable one. The software clears and re-issues the fault messages. When the IEEE-1570 was reconnected, the corresponding plug in failed to reinitialize.
	4.2	N/A	N/A No Preemption required	Normal	Run presented no errors.

Source: Battelle

Table 4 provides a summary list of the RCVW FT&E test cases. The table distinguishes test cases that are scenario-based versus those that are not, and specifies how the tested requirements are verified (I = inspect, T = test, D = demonstrate, and A = analyze) in each test case. Within the test cases, a vehicle type is referenced and is defined as follows:

- Type A = Passenger Car

Table 4. Summary of FT&E Test Cases

Test Case No.	Test Case Title	Phase III. Discrete Based FT&E Verification
1.1	CV Approaches and Travels through HRI, 35 mph, No HRI Active Signal	D, A
1.2	Type A CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions	D, A
1.3	Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions	D, A
1.4	Type A CV Approaches with Preemption Signal Detected at the Approach Midpoint, 35 mph, Dry Conditions	D,A
1.5	Type A CV Approaches with Preemption Signal Detected near the Approach Zone Hazard Zone Transition, 35 mph, Dry Conditions	D, A
1.6	Type A CV Approaches with Preemption Signal Detected at Approach Entry, Preemption signal is removed at Midpoint, 35 mph, Dry Conditions	D, A
2.1	CV Stopped in HRI Hazard Zone with Preemption Active/Not active	D,A
3.1	Loss of C-V2X after CV has received an RCVW	D,A
3.2	Loss of C-V2X Prior to entering the MAP	D,A
3.3	Loss of RTK Fix (GPS)	D,A
3.4	Loss of MAP	D,A
3.5	Loss of IEEE 1570 Heartbeat	D,A
4.1	Radio Range	D,A
4.2	GNSS Stability and Accuracy (Observation)	D

Source: Battelle

During testing, datasheets were used to manually record the following observed details for post-testing reconciliation and analysis with log files. The Battelle team was able to remotely connect to the RCVW VBS unit and continually monitor the system performance. For each test iteration, times were recorded in HH:MM:SS as taken from RCVW application internal clock as well as an on-hand clock. Information recorded included:

- Test start
- Test ending
- Any fault being issued during the test run
- General test observations

Table 5 summarizes test team personnel location and responsibilities, including data collection details. The control station was located near the roadside controller cabinet in which the RCVW RBS was integrated.

Table 5. Test Team Locations and Responsibilities for Scenario-Based FT&E Test Cases

Location	Responsibility	Action	Collection
CV	Driver	• Set entry speed using cruise control prior to Approach Zone entry • Take braking and steering action, as needed, to maintain safety	• Record notes following each test/test iteration
CV	Test Observer	• Collect data specified • Radio Coordination between CV and Roadside personnel • Monitor RCVW VBS system operation and ensure readiness for testing.	• Record Test Start time and End time, as well as time of noted events • Observe and record the time of issuing of inform and warning alerts for log reconciliation • Record events such as unanticipated of inform and warning alerts, or other behavior that may explain performance data in logs • Record any aborted tests and reason.
Roadside	Observer/Test Director	• Collect data specified • Radio coordination between tests times and preparations for next test • Control preemption signal activation/de-activation	• Record Test Start time and End time, as well as time of noted events • Record any aborted tests and reason • Upload log data from VBS and OBU as needed
Control Station	Algorithm Observer	• Monitor logging for errors and confirmation of what is happening within the RCVW algorithm. • Compile collected data from RBS, RSU, x-SPI, OBU, VBS and CBS on a periodic basis and backup • Lead or support reconciliation after test(s)	• Periodically upload log data from the RBS and CBS • Document and communicate to testing lead any issues needing resolution

Source: Battelle

FT&E Results Reconciliation Approach and Results Summary

RCVW data from two sources was used in the reconciliation of the FT&E results: 1) Transportation Message Exchange (TMX) core plugin logs from the VBS and 2) PCAP (packet capture) files of C-V2X communication details from the RBS and VBS C-V2X radios.

VBS TMX Core Log Based Analysis

VBS TMX core log files contain records of all state changes sent to the DVI from the RCVW plugin, including:

- Application Active / Inactive states – indicating that preemption is on or off when a CV is within the MAP area ingress lane (or Approach Zone)
- Rail Crossing Violation Warning issuance and clearance states – indicating when an RCVW warning is issued and cleared
- Vehicle Stopped at HRI Warning – issuance and clearance states - indicating when a 'Clear HRI' Warning is issued and cleared)
- Error activation and clearance messages (when an expected message is not received by the VBS from the RBS RSU, such as MAP or SPaT data, or when GPS communications are degraded)

TMX core log files additionally contain periodically recorded details from its input sources including location information from GNSS and J2735 messages from C-V2X. Other calculated values are used as position-based determinants in the inform/warning state changes described above. Some key recorded log details include:

- Latitude (GNSS)
- Longitude (GNSS)
- Heading (GNSS)
- RTK resolution type (GNSS: Fixed, Float, None)
- In-Lane Status (True/False)
- Speed (GPS)
- Preemption Status (HRI active signal, from J2735 SPaT message)
- Distance to the railroad crossing (calculated value by the RCVW application)
- Safety Stop Distance (calculated value by the RCVW application – distance determined to be needed given the current instantaneous conditions)

All TMX core log records are time stamped, with time expressed in Universal Time Code (UTC).

The following series of three TMX core log file excerpts are from FT&E, as noted.

Example 1 (Test 1.2): CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions. The log files show the system issuing an “Application Available” message on the DVI, followed by an Approach Inform Alert

```
tmx-Nuvo-7100VTC-Series c88afafe3d73[992]: [2024-07-25 20:32:41.034] Plugin/src/RCVWPlugin.cpp (1346) - DEBUG : CrossingDistance: 388.2994622, SafetyStopDistanceV1: 43.305805
tmx-Nuvo-7100VTC-Series c88afafe3d73[992]: [2024-07-25 20:32:41.035] Plugin/src/RCVWPlugin.cpp (1189) - DEBUG : Sending Application Message: Available
tmx-Nuvo-7100VTC-Series c88afafe3d73[992]: [2024-07-25 20:32:41.044] Plugin/src/RCVWPlugin.cpp (1201) - DEBUG : Sending Application Message: ApproachInform
tmx-Nuvo-7100VTC-Series c88afafe3d73[992]: [2024-07-25 20:32:41.230] WPlugin/src/RCVWPlugin.cpp (839) - DEBUG : LOC TIME, LOC SPEED, LOC HEADING, SPEED, HEADING, RTK, FREQUEN
```

The vehicle continues its approach at 35 mph and receives an Approach Warning as it nears the HRI.

```
tmx-Nuvo-7100VTC-Series c88afafe3d73[992]: [2024-07-25 20:32:58.422] Plugin/src/RCVWPlugin.cpp (1316) - DEBUG : Calculated Acceleration: -0.376344086, expectedStopDistance: 387.7609173
tmx-Nuvo-7100VTC-Series c88afafe3d73[992]: [2024-07-25 20:32:58.423] Plugin/src/RCVWPlugin.cpp (1346) - DEBUG : CrossingDistance: 96.00513651, SafetyStopDistanceV1: 53.54657296, SafetyStopDistance: 9
tmx-Nuvo-7100VTC-Series c88afafe3d73[992]: [2024-07-25 20:32:58.423] Plugin/src/RCVWPlugin.cpp (1213) - DEBUG : Sending Application Message: ApproachWarning
tmx-Nuvo-7100VTC-Series c88afafe3d73[992]: [2024-07-25 20:32:58.627] WPlugin/src/RCVWPlugin.cpp (839) - DEBUG : LOC TIME, LOC SPEED, LOC HEADING, SPEED, HEADING, RTK, FREQUENCY: 1721939578500, 17.001
```

Example 2 (Test 1.5) CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions. Example of the repeating System Fault message.

```
tmx-Nuvo-7100VTC-Series c88afafe3d73[968]: [2024-07-26 14:00:07.217] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: Location Data Not Received
tmx-Nuvo-7100VTC-Series c88afafe3d73[968]: [2024-07-26 14:00:07.587] WPlugin/src/RCVWPlugin.cpp (839) - DEBUG : LOC TIME, LOC SPEED, LOC HEADING, SPEED, HEADING, RTK, FREQUENCY: 17220024
tmx-Nuvo-7100VTC-Series c88afafe3d73[968]: [2024-07-26 14:00:07.629] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error
tmx-Nuvo-7100VTC-Series c88afafe3d73[968]: [2024-07-26 14:00:07.649] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: Location Data Not Received
tmx-Nuvo-7100VTC-Series c88afafe3d73[968]: [2024-07-26 14:00:07.938] WPlugin/src/RCVWPlugin.cpp (839) - DEBUG : LOC TIME, LOC SPEED, LOC HEADING, SPEED, HEADING, RTK, FREQUENCY: 17220024
tmx-Nuvo-7100VTC-Series c88afafe3d73[968]: [2024-07-26 14:00:07.959] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error
```

Example 3 (Test 2.1): CV Stopped in HRI Hazard Zone. The log file shows the system issuing an “HRI Warning” Message on the DVI.

```
tmx-Nuvo-7100VTC-Series c88afafe3d73[992]: [2024-07-25 20:40:16.535] Plugin/src/RCVWPlugin.cpp (1316) - DEBUG : Calculated Acceleration: -0.8730964467, expe
tmx-Nuvo-7100VTC-Series c88afafe3d73[992]: [2024-07-25 20:40:16.536] Plugin/src/RCVWPlugin.cpp (1346) - DEBUG : CrossingDistance: -1, SafetyStopDistanceV1:
tmx-Nuvo-7100VTC-Series c88afafe3d73[992]: [2024-07-25 20:40:16.536] Plugin/src/RCVWPlugin.cpp (1225) - DEBUG : Sending Application Message: HRIWarning
tmx-Nuvo-7100VTC-Series c88afafe3d73[992]: [2024-07-25 20:40:16.721] WPlugin/src/RCVWPlugin.cpp (839) - DEBUG : LOC TIME, LOC SPEED, LOC HEADING, SPEED, HEA
tmx-Nuvo-7100VTC-Series c88afafe3d73[992]: [2024-07-25 20:40:16.727] Plugin/src/RCVWPlugin.cpp (1013) - DEBUG : Lane, SignalGroup = 0, -1
```


Reconciling the VBS TMX core log files included the following observations:

- All expected state changes are logged and no unexpected state changes were seen, based on the test case scenario executed
- State changes logged are appropriate relative to calculated or reported TMX core log values (e.g., “HRI Warning” is issued when vehicle speed is zero, vehicle is within the HRI, and “ApproachWarning” is issued when calculated safety stop distance exceeds calculated crossing distance and clears when crossing distance exceeds safety stop distance or when vehicle crosses into the HRI or when preemption state is false, etc.)

Paquet Capture Log Files Based Analysis

PCAP files contain the J2735 SPaT and MAP messages being sent and received from both the roadside and vehicle-based C-V2X radios are used to analyze the latency requirements of the system. All PCAP messages logged provide the following:

- A timestamp (sent and received depending on the source)
- A message length
- A data frame that includes message ID, intersection geometry (for MAP messages) or intersection state (for the SPaT messages) which includes preemption information.

The following (see Figure 1010) is an example of the PCAP file from the Roadside Radio (transmitting) for the last day of testing.

No.	Time	Length	Data
1	2024-07-26 14:13:12.602083	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a6e6020...
2	2024-07-26 14:13:12.694679	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a68b8020...
3	2024-07-26 14:13:12.810861	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a6992020...
4	2024-07-26 14:13:12.913836	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a6a68020...
5	2024-07-26 14:13:13.015694	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a6b30020...
6	2024-07-26 14:13:13.115077	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a6bfa020...
7	2024-07-26 14:13:13.219113	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a6cd2020...
8	2024-07-26 14:13:13.330381	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a6da8020...
9	2024-07-26 14:13:13.428664	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a6e82020...
10	2024-07-26 14:13:13.486841	128	01030080073a0380373e0001060a4d75de4839ab360e19b204e00000160901080b0a53524d536572766963658002...
11	2024-07-26 14:13:13.533313	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a6f50020...
12	2024-07-26 14:13:13.570106	316	010300c2f92880f4038081f0001280ec380530203073be154d75de4439ab3522198c02dc051f702b2009707870e5...
13	2024-07-26 14:13:13.659386	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a702c020...
14	2024-07-26 14:13:13.762879	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a7108020...
15	2024-07-26 14:13:13.869995	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a71e2020...
16	2024-07-26 14:13:13.989041	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a72bc020...
17	2024-07-26 14:13:14.078159	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a7384020...
18	2024-07-26 14:13:14.186685	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366cbcb9efdc39df02001091f6a7462020...

Figure 10. PCAP File Example – (Source: Battelle)

The protocol analysis tool called Wireshark was used to export the necessary data from the PCAP files into an Excel spreadsheet format. The data was then filtered to extract only the MAP and SPaT data that was sent by the roadside unit and received by the vehicle-based radio for each of the test run iterations. Filtering was done via the message length attribute, SPaT and MAP message lengths are 125 and 316, respectively.

Figure 11 is an example of the analyzed SPaT Frequency Analysis. The timestamps were used to determine how much time passed between each message. The detailed results of the analysis can be found in [DSRC Messaging and Data Processing Latency Requirement Verification](#).

Summary of RCVW Field Test and Evaluation Approach Results

No.	Time	Length	Data	Time since last
1	7/26/2024 10:13:12.602	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.0000
2	7/26/2024 10:13:12.695	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.0930
3	7/26/2024 10:13:12.811	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1160
4	7/26/2024 10:13:12.914	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1030
5	7/26/2024 10:13:13.016	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1020
6	7/26/2024 10:13:13.115	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.0990
7	7/26/2024 10:13:13.219	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1040
8	7/26/2024 10:13:13.320	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1110
9	7/26/2024 10:13:13.429	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.0990
11	7/26/2024 10:13:13.533	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1040
13	7/26/2024 10:13:13.659	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1260
14	7/26/2024 10:13:13.763	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1040
15	7/26/2024 10:13:13.870	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1070
16	7/26/2024 10:13:13.989	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1190
17	7/26/2024 10:13:14.078	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.0890
18	7/26/2024 10:13:14.187	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1090
19	7/26/2024 10:13:14.302	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1150
20	7/26/2024 10:13:14.410	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1080
22	7/26/2024 10:13:14.504	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.0940
24	7/26/2024 10:13:14.603	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.0990
25	7/26/2024 10:13:14.708	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1050
26	7/26/2024 10:13:14.814	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1060
27	7/26/2024 10:13:14.939	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1250
28	7/26/2024 10:13:15.051	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1120
29	7/26/2024 10:13:15.175	125	01030080023703803400133100385a161e9d32ecd995057cbcf209597366c	0.1240

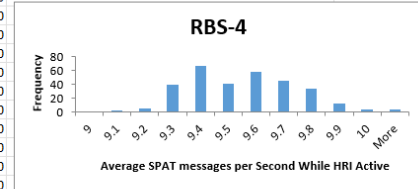


Figure 11. SPaT message frequency analysis

A similar analysis was performed for MAP message frequency (see Figure 12). The detailed results of the analysis can be found in C-V2X Messaging and Data Processing Latency Requirement Verification.

No.	Time	Length	Data	Time since last
12	7/26/2024 10:13:13.570	316	010300c2f92880f4038081f0001280ec3f	0
23	7/26/2024 10:13:14.542	316	010300c2f92880f4038081f0001280ec3f	00.972
34	7/26/2024 10:13:15.554	316	010300c2f92880f4038081f0001280ec3f	01.012
46	7/26/2024 10:13:16.555	316	010300c2f92880f4038081f0001280ec3f	01.001
57	7/26/2024 10:13:17.563	316	010300c2f92880f4038081f0001280ec3f	01.008
69	7/26/2024 10:13:18.587	316	010300c2f92880f4038081f0001280ec3f	01.024
80	7/26/2024 10:13:19.578	316	010300c2f92880f4038081f0001280ec3f	00.991
92	7/26/2024 10:13:20.603	316	010300c2f92880f4038081f0001280ec3f	01.025
103	7/26/2024 10:13:21.600	316	010300c2f92880f4038081f0001280ec3f	00.997
115	7/26/2024 10:13:22.601	316	010300c2f92880f4038081f0001280ec3f	01.001
127	7/26/2024 10:13:23.601	316	010300c2f92880f4038081f0001280ec3f	01.000
138	7/26/2024 10:13:24.632	316	010300c2f92880f4038081f0001280ec3f	01.031
150	7/26/2024 10:13:25.611	316	010300c2f92880f4038081f0001280ec3f	00.979
161	7/26/2024 10:13:26.627	316	010300c2f92880f4038081f0001280ec3f	01.016
173	7/26/2024 10:13:27.611	316	010300c2f92880f4038081f0001280ec3f	00.984
184	7/26/2024 10:13:28.612	316	010300c2f92880f4038081f0001280ec3f	01.001
196	7/26/2024 10:13:29.628	316	010300c2f92880f4038081f0001280ec3f	01.016
208	7/26/2024 10:13:30.648	316	010300c2f92880f4038081f0001280ec3f	01.020
219	7/26/2024 10:13:31.655	316	010300c2f92880f4038081f0001280ec3f	01.007
231	7/26/2024 10:13:32.656	316	010300c2f92880f4038081f0001280ec3f	01.001
242	7/26/2024 10:13:33.638	316	010300c2f92880f4038081f0001280ec3f	00.982
254	7/26/2024 10:13:34.645	316	010300c2f92880f4038081f0001280ec3f	01.007
265	7/26/2024 10:13:35.672	316	010300c2f92880f4038081f0001280ec3f	01.027
277	7/26/2024 10:13:36.659	316	010300c2f92880f4038081f0001280ec3f	00.987
289	7/26/2024 10:13:37.663	316	010300c2f92880f4038081f0001280ec3f	01.004

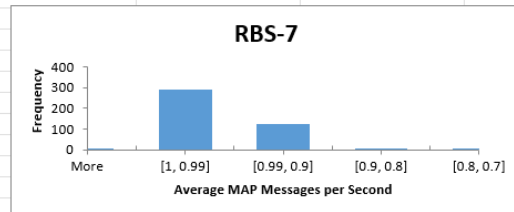


Figure 12. MAP message frequency analysis

Summary of RCVW Performance

A detailed account of all FT&E test results can be found in the next chapter, and a summary of performance by requirement can be found in Appendix.

RCVW Requirements Traceability Matrix. In summary all requirements were confirmed to have been met with the following exceptions: 1) requirements RBS-004, RBS-022, and VBS-022 were not confirmed (see C-V2X Messaging and Data Processing Latency Requirement Verification).

Across the scenarios focused on assessing violation detections with the CV on approach (i.e., Test Cases 1.2 – 1.6) the RCVW system was found reliably warn the driver when the vehicle was approaching the active crossing at a speed calculated to be too fast for the stopping distance (using the calculations and parameters described above). Following the actions of the vehicle driver, the vehicle was able to safely decelerate and stop before reaching the crossing.

System Performance

The following is a summary of the system performance where distinct factors were entered into the test to verify and validate the requirements. All the expected inform and warning alerts were issued to the driver as expected. Log files were reviewed to ensure that the correct message was issued and to verify that no unseen alerts were created.

Test cases 1.3-1.5 were performed with the train preemption signal issued after the CV had entered the approach. The purpose of these tests was to understand the system performance, verify the AASHTO Green Book Stopping Sight Distance formula used and validate if the CV would have enough distance to reach a full stop before the stop bars. The vehicle speed during these tests was 35 mph. These test cases were designed to issue the preemption activation at the 3/4-mark, midpoint, and 1/4 mark of the approach. The vehicle had ample time to travel down the approach lane, receive the warning alert, safely decelerate, and come to a complete stop before reaching the stop bar.

In Test Case 1.6, the preemption signal was removed at the midpoint after the vehicle had entered the approach. An HRI Active Inform Alert was issued as soon as the vehicle entered the MAP. The train preemption was removed before a warning alert message was issued, removing the inform alert and allowing the vehicle to travel across the HRI with no nuisance alerts.

Test case 2.1 was designed to demonstrate the in-vehicle warning alerts the system would produce when the CV is stopped within the HRI. The system is designed to present a “clear track” warning when a vehicle is stationary at the HRI and a more emphatic one if the warning devices become active. The verification of this capability was not fully successful. See Summary of Detected Issues and Anomalies and Potential Refinement for more information

When the reception rate of essential information drops below a predetermined value or the data is not received, the system is designed to generate a System Fault warning message. The correct fault message was issued for every programmed fault detection. See System Fault Verification for more information

GNSS Position

The GNSS module in the RBS subsystem produces RTCM corrections which are broadcast via C-V2X. These RTCM corrections contain real-time data about the GNSS network, as well as perturbations in the ionosphere and troposphere which are used by the VBS GNSS module to correct the position solution of the vehicle and obtain an RTK position solution (sub-decimeter level positional accuracy). Issues with the reception and decoding of RTCM messages at the VBS resulted in the vehicle not being able to reach an RTK position solution. Without the processing of RTCM corrections, the GNSS device was only able to reach a sub-meter level positional accuracy. See C-V2X Messaging and Data Processing Latency Requirement Verification section for more information.

System Stability When Near/In/Beyond the HRI

A test case was designed to validate the stability of the GNSS System by confirming the following:

- A Clear HRI alert is displayed and remains when the vehicle is positioned within the HRI and remains in the same state as the vehicle remains static in the location.

The test case verifies that a system available is displayed when the vehicle is positioned at the edge of the HRI and remains in the same state as the vehicle remains static in the location. It also confirms that a Clear HRI is issued when the vehicle or a section of the vehicle (front or back) is positioned within the HRI.

These tests were conducted to confirm that a driver that stops along the boundary of the HRI Hazard Zone does not receive nuisance/intermittent warning/lack of warning alerts (i.e., the GPS accuracy is sufficient for the VBS to reliably know where the vehicle is located relative to the HRI Hazard Zone boundaries, and this fix does not drift over a significant period). The test was performed at each of the locations shown below:

- a) Vehicle is within Approach Zone, just outside of the HRI Hazard Zone. The front of the vehicle is right at the edge of the Hazard Zone.
- b) Front of Vehicle is inside the HRI Hazard Zone and rear of the vehicle is in the Approach Zone.
- c) Entire Vehicle is inside the HRI Hazard Zone
- d) Front of Vehicle is beyond the HRI Hazard Zone and rear of the vehicle is inside the HRI Hazard Zone.
- e) Entire vehicle is beyond (outside of) the HRI Hazard Zone. The back of the vehicle is just at the edge of the Hazard Zone.

In all locations the system produced the correct warning alerts when necessary and there was no nuisance/intermittent warning or lack of warning alerts. Due to the unavailability of processing RTCM corrections, the vehicle location accuracy was within one meter. This resulted in lack of repeatability when conducting iterations of the test case. On each iteration, the vehicle location was off by a few decimeters when compared to the previous iteration and the warning alerts were received in different physical locations. However, no nuisance/intermittent warning or lack of warning alerts were experienced on each test iteration.

System Fault Verification

Specific test cases were designed and performed to verify and validate the different system faults the system recognizes.

1. Loss of C-V2X after the CV has received an RCVW
2. Loss of C-V2X before the CV enters the MAP
3. Loss of RTK Fix (GPS)
4. Loss of MAP
5. Loss of IEEE 1570 Heartbeat

The loss of C-V2X was replicated by disabling the MessageReceiver plugin on the VBS after the issue of a warning alert for case 1, and prior to the entrance of the vehicle to the approach for case 2. For loss of MAP, the RBS MAP plugin was disabled. The loss of the IEEE 1570 Heartbeat was replicated by disconnecting the IEEE 1570 serial interface controller to the roadside CPU, effectively suspending the reception of the Heartbeat. For the loss of RTK Fix, it was planned for the broadcast of RTCM corrections to be disabled once the vehicle received a warning alert. Due to issues with the VBS not correctly processing RTCM corrections, the disabling of the RTCM broadcast was not required. See Summary of Detected Issues and Anomalies and Potential Refinement for more information

The system effectively issued a System Fault alert for all the conditions being tested. The log files were analyzed to verify that the correct fault alert was diagnosed and issued.

Radio Range

The replacement of the DSRC radios with C-V2X presented an opportunity to test the range of radio communications. The RCVW is designed with a parameter called “Distance to HRI”. This parameter provides a distance radius from a known HRI for the VBS to start the processing of MAP and SPaT. The default value for this parameter is 480 meters. The test required to update this parameter to 1 kilometer (km) to allow for this check to occur at a higher distance between the vehicle and the RBS. Two approaches that allowed the vehicle to travel towards the RBS beyond the 1 km radius were selected. An adjacent road was used where the vehicle traveled northbound and southbound towards the RBS. A third location was specifically chosen to include buildings and natural obstructions to observe radio connectivity.

The VBS was expected to look for MAP and SPaT from the HRI during each test since it would be located within the radius set. When MAP and/or SPaT signal is not received, the VBI would issue a system fault. When the signal is received, the VBS would issue a System Ready message. This would identify the locations where the communication signal was lost and regained.

During the test event, the “Distance to HRI” parameter was not updated. This resulted in the RCVW not issuing faults or events. However, the algorithm observer was able to detect signal receptions and losses and informed the rest of the team. Time records for each of these were captured and used to extrapolate to the VBS log files to detect the communication loss/gain latitude and longitude. This data was then used to calculate communication ranges.

Southbound Approach

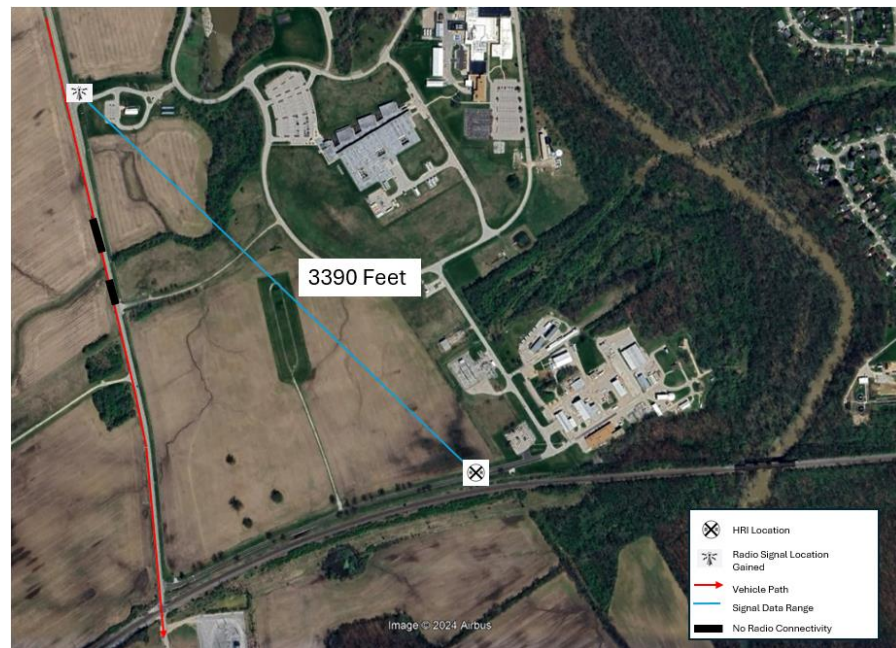


Figure 13. Southbound Approach (Source: Google Maps)

The VBS was able to process RBS signals at a distance of around 3390 ft (1.033 km). As the vehicle continued the travel south, communications were lost on two occasions. On both of these, it is suspected that the presence of vegetation obstructed the field of view of the vehicle antenna and the RBS antenna.



Figure 14 Southbound Approach – 1st Signal Loss (Source: Google Maps)

Figure 14 shows the location of the first signal loss during the southbound approach. It is observed that a tree and foliage exist on the left side of the approach, directly blocking line of sight to the RBS.



Figure 15. Southbound Approach – 2nd Signal Loss (Source: Google Maps)

Figure 15 shows a street view of the location where the communications between the VBS and RBS were lost during the southbound approach. Again, large trees and vegetation obstruct the LOS of the RBS.

Northbound Approach

On the northbound approach (See Figure 16), radio communications were not gained until the vehicle was within a perpendicular line of sight to the RBS. The radio range in this case was around 2000 ft (609.6 m).



Figure 16. Northbound Approach (Source: Google Maps)

The line of sight from this approach has several obstructions:

- Large Wooded area. See the red box in Figure 17
- Electric Power Plant. See the blue box in Figure 17



Figure 17. Northbound Obstructions (Source: Google Maps)

It is suspected at first the signal was blocked by the trees located both in the wooded area as well as along the south side of the test track. Once the vehicle cleared the wooded area, it is possible that the adjacent power plant caused interference with the signal.

Urban Setting Approach

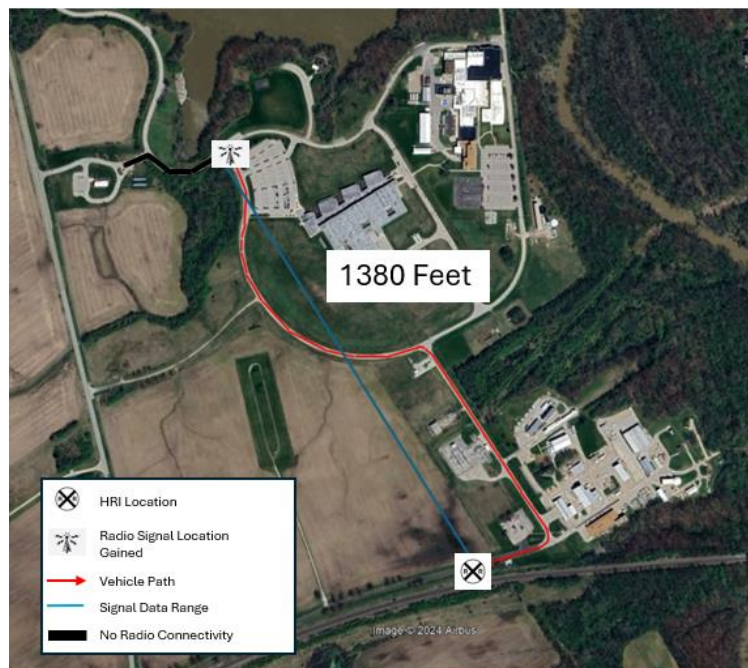


Figure 18. Urban Setting Approach (Source: Google Maps)

The urban setting approach was designed to have the vehicle approach the RBS using roads where buildings and vegetation were present. Figure 18 shows how radio connection was achieved once the vehicle cleared a wooded area on its right and was able to have LOS of the RBS. This represents a distance of 1380 ft (420.6 m). The vehicle proceeded to approach the crossing and radio connectivity remained throughout.

Due to the location of the test track, a test case where we would verify an obstacle-free radio communications range was not possible to be performed.

C-V2X Messaging and Data Processing Latency Requirement Verification

The following requirements are associated with the RCVW system latency. All requirements were verified and validated following the process described in the Paquet Capture Log Files Based Analysis.

- RBS-004 – The RBS shall broadcast the HRI Activation Status message to both the VBS and CBS 10 times per second during operation.
- RBS-007 – The RBS shall broadcast the HRI Geometry to both the VBS and CBS once per second during operation.
- VBS-022 – The VBS shall be capable of receiving messages sent by the RBS within 50 ms.
- VBS-023 – The VBS shall be capable of processing received data within 85 ms.

Log data from 7/26/2024 was analyzed and the rate of broadcasted SPaT and MAP data was calculated. SPaT messages were analyzed by collecting all HRI active messages throughout the day, separating them by the individual seconds they were sent, and then averaging how many messages each second contained. MAP messages were analyzed via the same process,

but the messages were separated into individual minutes instead. The RBS broadcasted on average 9.545 SPaT messages and 1 MAP message per second.

The lowest rate of MAP messages broadcasted during a full minute (the minute was not cut off due to the beginning or end of the log file) was 0.9833 messages per second (59 messages over the minute) and the highest was 1 message per second (60 messages over the full minute). The lowest rate of SPaT messages broadcasted during a full second (the second was not cut off due to the beginning or end of the log file) was 9.01 messages per second and the highest was 10.72 messages per second.

A MAP message is meant to be sent at a rate of 1 Hz (1 message per second). The worst-case scenario seen in the analysis is that 1 message was missed in the minute ($1/60 = 1.7\%$). For the case of SPaT messages, the broadcast rate is 10 Hz (10 messages per second). The worst-case scenario in the analysis was that 1 message was missed per second or around 60 over a minute (59.4 to be precise). The total SPaT lost was ($60/600 = 1.0\%$).

This is not considered to be a very high variation on the data received.

Figure 19 and Figure 20 show the distribution of the message rates for MAP and SPaT respectively.

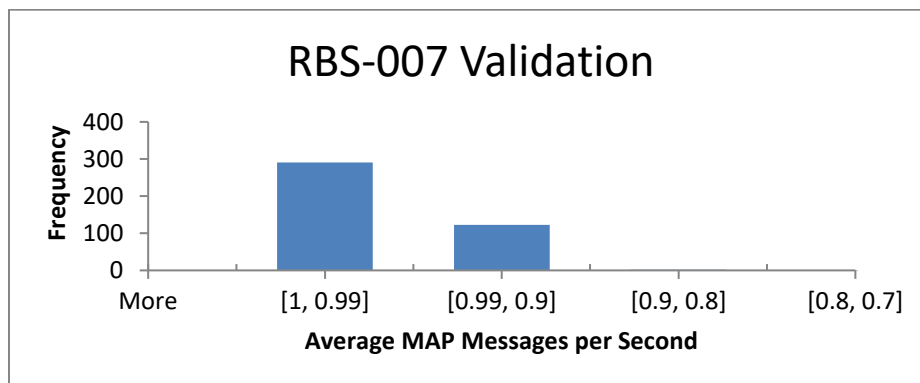


Figure 19. Average MAP Message Sent by the RBS Per Second (Source: Battelle)

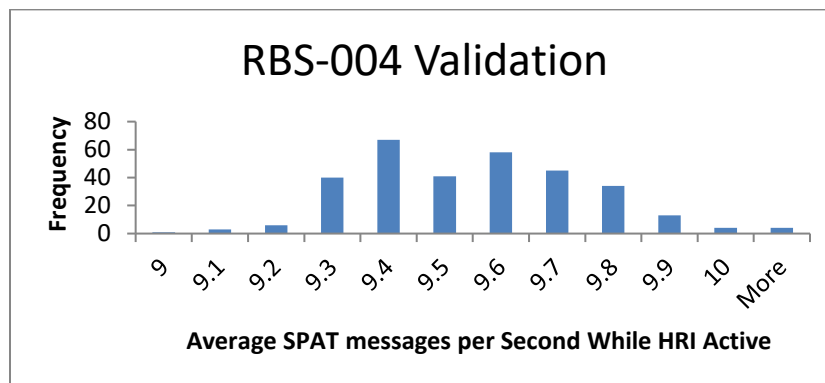


Figure 20. Average SPaT Message Sent by the RBS Per Second (Source: Battelle)

While the analysis shows the system meets requirement RBS-007, it fails to meet requirement RBS-004 by not achieving a rate of 10 SPaT messages per second.

VBS-022 requires the reception of a roadside message within 50 ms. after being sent. The procedure followed for verification extracted the timestamps of messages sent by the RBS and then compare the difference to the timestamp of when it was received by the VBS. The messages were also split between the MAP and SPaT variants. This approach allowed an analysis of the two main messages being exchanged by the two subsystems. A discrepancy between the clocks on the RBS and VBS was noticed, meaning that the timestamps between the messages being sent and received had no correlation and resulted in not being able to perform this analysis. This issue is further explained in RBS/VBS Time Discrepancies.

It should be noted that the results of the analysis in the previous phase (RVCW-II) showed that the vehicle-based subsystem was capable of receiving messages sent by the roadside-based subsystem within 50 ms.

Verification of Requirement VBS-023

VBS-023 requires the system to be capable of processing received data within 85 ms. The data received is directly processed by the RCVW plugin, this data comes asynchronously and is processed every time it receives a location (GNSS) message which is generated by the Location plugin. The analysis to validate this requirement looks at the processing time between receiving a location message and generating an alert as a result of that message. Tests 1.2-1.6 were analyzed by extracting the time when the RCVW log file indicates an Approach Inform or Approach Warning were issued and the time the latest location message was received. The difference between these two times indicates how long it takes RCVW to fully process the data and generate an alert.

Data from the field test log files does not allow for an end-to-end latency analysis. Additional bench tests are needed to determine the time required for the Location plugin to decode and process location information. Currently, the log files only provide the time the Location plugin delivers the already processed location data. Adding this to the processing time already described would provide a complete latency analysis (focused only on GPS processing time). One more thing to consider is that the system does not perform calculations when every piece of data arrives. This is because this data arrives asynchronously, and location data is considered the most important. MAP, SPaT, and VBM messages are cached when received and the last one of each is used when a location message is received. Since location messages are received at ~10 Hz, it is possible that all the other cached data could be up to 100 ms. latent when calculations are performed.

The analysis shows that on average the processing time after a location message is received by the RCVW plugin is 12 milliseconds. The maximum latency observed was 22 milliseconds and the minimum was 4 milliseconds. It was also observed that the system was far more efficient in processing Approach Warnings, averaging 7.4 milliseconds, than Approach Informs, averaging 17 milliseconds.

There is not much control the code has over what happens across a multi-threaded application such as this, so it is difficult to say why there is the average 56% discrepancy in processing time for effectively making the same decision. The code could be locking itself out for a number of milliseconds, or simply being blocked by execution of another section of the code. The check for the Inform case (which takes longer on average) occurs prior to the check for the Warning case. It could be argued that it is more correct to check for the Warning condition first because it will always supersede the Inform, meaning that if the Warning is active, a check for the Inform is unnecessary and wasteful. However, on the surface it is unlikely that swapping the order of the checks would affect the average time, but it is a possibility.

Even more, both of these notices (and all the other Faults and Alerts for that matter) do something that negatively affect the average performance of that operation in that they update the plugin “status” before sending the application message. Updating the status entails caching the new value and then sending a status update message from the plugin to any listeners. This is problematic because the more urgent application message that triggers to the UI update must wait until the status message is sent. It is possible that this could block the sending (and thus receiving) of the application message for a few milliseconds. It seems smarter to do the less critical “status” update after the UI message has been sent instead of the other way around. Perhaps by removing this variable-length operation prior to performing a critical event would balance out the average time it takes to send each critical event.

We will mark it as a known issue and write a bug ticket for future resolution,

Figure 21 shows the distribution of the latency data for tests 1.2-1.6.

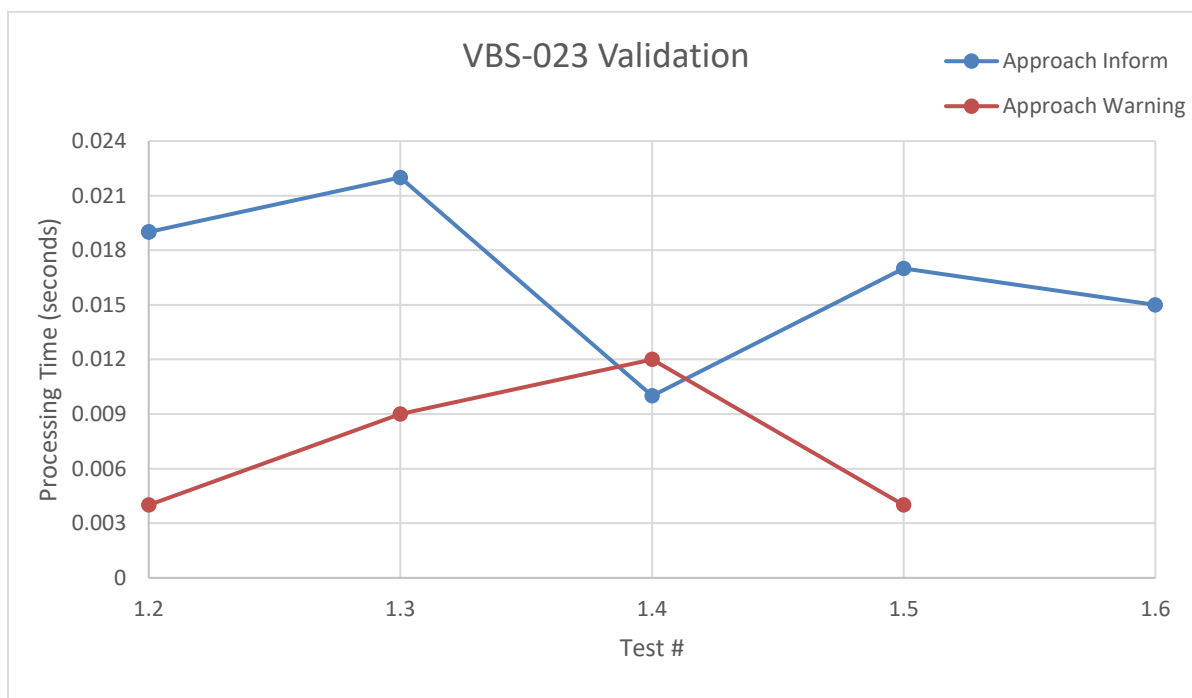


Figure 21. Processing Latency (Source: Battelle)

The analysis shows that the processing latency for tests 1.2-1.6 was well less than the 85 ms. required by VBS-23.

Verification of Requirement RBS-022.

Requirement RBS-022 states the following:

“The RBS shall be able to produce RTCM messages directly through the use of a local base station.”

To comply with this requirement, the RCVW RBS is designed to generate and broadcast RTCM corrections by using a base station as part of the system. These corrections are broadcasted over C-V2X radio for any incoming CV to acquire and use to generate an RTK location solution.

During the field test event, issues with the reception of this RTCM corrections by the VBS were experienced. The RBS sends RTCM messages over C-V2X, which are to be applied to the receiver on the VBS. While there was no specific indication that anything was wrong with the

RTCM messages themselves, they were not being applied to the receiver because no RTK fix was achieved. This issue could have been caused by incorrect settings on the receiver, invalid messages, or a failure to write the bytes to the receiver correctly. To date, no attempt has been made to determine the specific cause of the failure during the field test. Instead, the GNSS receiver was altered to operate under satellite-based augmentation (SBAS) and the system was configured to ignore the RTK validation check during field test.

The RCVW VBS is designed to check the incoming rate of RTCM messages. When these messages fall below a predetermined rate (programmable via a system parameter), a system error is issued and displayed via the system DVI. To be able to continue with the test event, it was decided to disable the check for RTCM corrections. However, this check was enabled for test case 3.3 Loss of RTK Fix (GPS).

The issue is being investigated.

Summary of Detected Issues and Anomalies and Potential Refinement

The following detected issues and anomalies, and potential refinement opportunities were identified during FT&E. All the issues and anomalies will be corrected and retested prior to the end of the project. This document will be updated to reflect the results.

Flickering Fault

Test cases 3.1 to 3.5 were designed to verify the correct issuing of system faults when critical data was not received or received at a different rate than what is required. An issue was observed when a fault was generated by the RCVW system. When the necessary fault message was generated, the system immediately being cleared this fault message. A few seconds later, the error was reissued and once again, cleared by the system. This continued until the fault was resolved cleared.

Figure 22 shows an extract of the log files which present the system error as a result of missing C-V2X data (in this particular case SPaT messages). The log files show the sending of an Application Message : Error SPaT at 14:16:58.172. Around 100 ms. later, the error was cleared by the system. Around 2 seconds later the error was reissued, only to be cleared once again. This was a cyclical event that continued throughout the test case. It is important to note, however, that this behavior only occurs for the fault events, and not for any other alerts.

```
Jul-26-10:16:58-tmx-Nuvo-7100VTC-Series-c88afafe3d73[968]:[2024-07-26-14:16:58.172].  
Plugin/src/RCVWPlugin.cpp-(1240)---DEBUG--:Sending-Application-Message:Error:SPAT-  
Data-Not-Received¶  
Jul-26-10:16:58-tmx-Nuvo-7100VTC-Series-c88afafe3d73[968]:[2024-07-26-14:16:58.282].  
Plugin/src/RCVWPlugin.cpp-(1246)---DEBUG--:Sending-Application-Message:Clear-Error¶  
Jul-26-10:17:00-tmx-Nuvo-7100VTC-Series-c88afafe3d73[968]:[2024-07-26-14:17:00.215].  
Plugin/src/RCVWPlugin.cpp-(1240)---DEBUG--:Sending-Application-Message:Error:SPAT-  
Data-Not-Received¶  
Jul-26-10:17:00-tmx-Nuvo-7100VTC-Series-c88afafe3d73[968]:[2024-07-26-14:17:00.326].  
Plugin/src/RCVWPlugin.cpp-(1246)---DEBUG--:Sending-Application-Message:Clear-Error¶  
Jul-26-10:17:02-tmx-Nuvo-7100VTC-Series-c88afafe3d73[968]:[2024-07-26-14:17:02.259].  
Plugin/src/RCVWPlugin.cpp-(1240)---DEBUG--:Sending-Application-Message:Error:SPAT-  
Data-Not-Received¶  
Jul-26-10:17:02-tmx-Nuvo-7100VTC-Series-c88afafe3d73[968]:[2024-07-26-14:17:02.369].  
Plugin/src/RCVWPlugin.cpp-(1246)---DEBUG--:Sending-Application-Message:Clear-Error¶
```

Figure 22. SPaT Data System Fault and Clear Log (Source: Battelle)

This is suspected to be a software logic error that is currently being investigated.

Vehicle in HRI Warning vs Inform Alerts

RCVW includes a capability where the system issues two different alerts according to the position of the vehicle within the HRI. As a CV enters the HRI and stops, it checks if HRI preemption is ON or OFF and depending on this state, the system will produce an inform alert (preemption off) or a warning alert (preemption on). These messages are to encourage the driver to evacuate the HRI as soon as possible. Issues were experienced when testing this functionality.

The analyzed log files show that with train preemption OFF and when the vehicle enters the HRI and stops within the HRI boundaries, the system detects this and issues an HRI Warning application message. This is not the correct behavior as it should have issued an HRI Inform alert. (see Figure 23). The issue is still being investigated.

```
[2024-07-25 20:44:14.050] Plugin/src/RCVWPlugin.cpp (1283) - DEBUG : Latitude: 39.9570093, Longitude: -83.2478238, Speed: 0.714, PrevSpeed: 1.043, HDOP: 0, Preemption: 0
[2024-07-25 20:44:14.051] Plugin/src/RCVWPlugin.cpp (1346) - DEBUG : CrossingDistance: -1, SafetyStopDistanceV1: 1.11976875, SafetyStopDistance: 7.542589139, ExpectedStopDistance:
[2024-07-25 20:44:14.051] Plugin/src/RCVWPlugin.cpp (1225) - DEBUG : Sending Application Message: HRI Warning
```

Figure 23. Vehicle In HRI Warning log file (Source: Battelle)

RBS/VBS Time Discrepancies

To determine a clear time of travel between radio transmission and radio reception, the clocks at both the transmitter and receiver must be aligned. The typical solution is to utilize a time synchronization tool (such as chrony on Linux) that keeps the local time of the computing system up to date with that of a source, such as a GPS signal or a time server. Much care was taken during the field test to ensure the clocks on the RBS and the VBS utilized chrony from several available sources so that the clocks were accurately synchronized. However, the out-of-the-box chrony implementations on the RSU and OBU radios themselves were not formally verified before testing began. Furthermore, while some chrony tools exist to show the correct application of the time sources and their respective estimated skew, there is no straightforward way to confirm the difference in time between two distinct systems. Therefore, while the data exists on the radio logs determining the point in time at which the message was sent and received, the difference between those two times is not a reliable measurement for the latency since either one or both radio clocks may have been off real-time by a significant amount. No other data points were distinguishable from the VBS and RBS computing logs that would indicate a consistent time differential between each other or between the radios. Thus, there can be no deterministic conclusion made for the travel time requirements of the C-V2X radio without further testing.

Configuration Issues

The following issues occurred during field testing as a result of wrong configuration of the system.

- 100% of the RCVW RBS logs for the first day of testing were lost.
- Issues with the display of the In-HRI Alert on the System DVI were observed.
- Ubuntu system requests for version updates.

All issues have been identified and corrected. A Standard Operating Procedure is being produced which will detail the correct configuration procedures for the RCVW system. This document will be delivered as part of the project deliveries.

Regression Testing

Several issues and anomalies were identified during the Field Test activities (See Summary of Detected Issues and Anomalies and Potential Refinement). The development team analyzed each issue and implemented several software code corrections to ensure that all expected requirements and capabilities are met.

A regression Field Test activity was performed on December 12 and 13, 2024 at the West Jeff location where specific test cases were selected.

Test Case No.	Test Case Title
1.2	Type A CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions
1.3	Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions
2.1	CV Stopped in HRI Hazard Zone with Preemption Active/Not active
3.1	Loss of C-V2X after CV has received an RCVW
3.3	Loss of RTK Fix (GPS)
3.4	Loss of MAP
3.5	Loss of IEEE 1570 Heartbeat
4.2	GNSS Stability and Accuracy (Observation)

These test cases were selected to ensure that the following is verified:

- Confirm the reception and application of RTCM corrections in the VBS.
- RCVW system being capable of producing distinct alerts when a vehicle is within the HRI and preemption signal is On/Off.
- Stability of the GNSS module when RTK corrections have been applied.

Test Case Number	CV Location at Preemption Signal Detection	CV Speed at Approach Entry	RCVW State
1.2	Start of Approach	35 mph	Normal
1.3	$\frac{3}{4}$ point	35 mph	Normal
2.1	Stopped on Tracks (preemption will deactivate during the test)	N/A	Normal
3.1	Start of Approach	35 mph	Fault – C-V2X
3.3	Start of Approach	35 mph	Fault – GPS
3.4	Start of Approach	35 mph	Fault – MAP
3.5	Start of Approach	35 mph	Fault – IEEE-1570
4.2	N/A No Preemption Required	N/A	Normal

In addition, the cloud-based subsystem was integrated into the test location to verify CBS related requirements with a complete RCVW system.

Figure 24 shows the inclusion of the CBS to the existing RBS architecture.

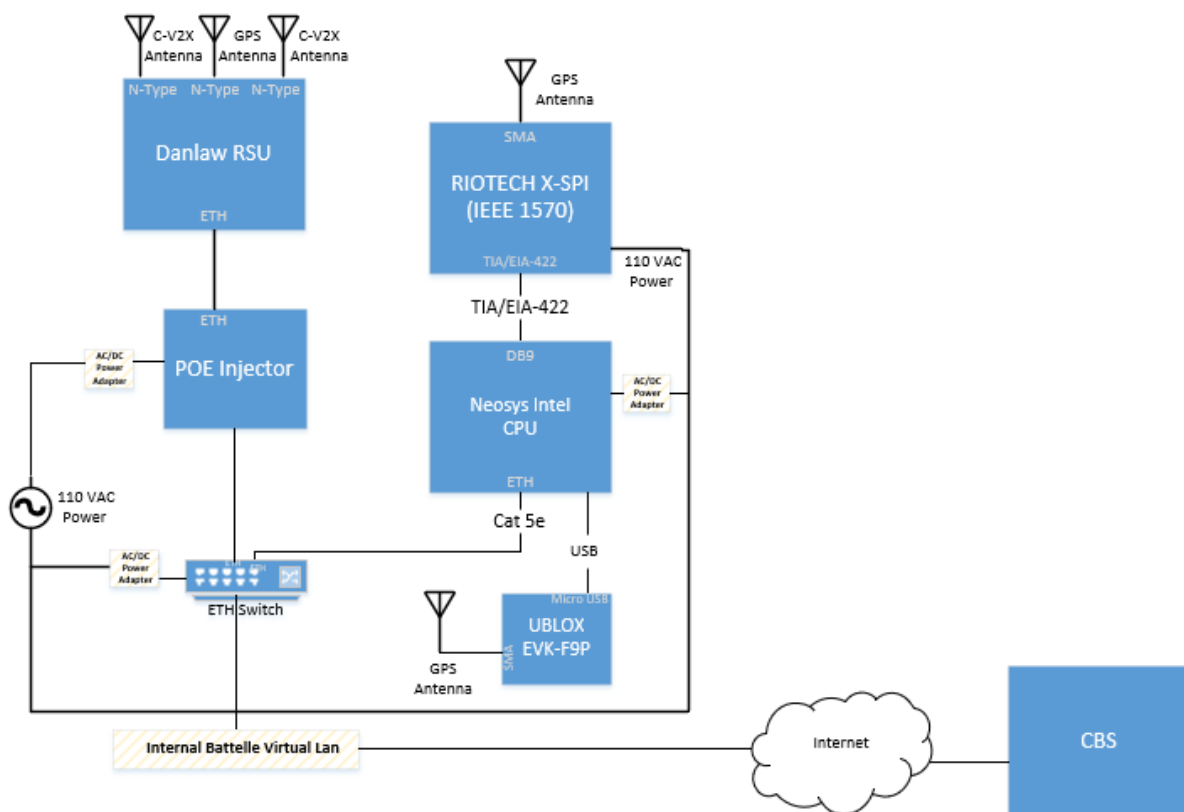


Figure 24. Complete RCVW System

The regression field test focused (for the case of the CBS) on the verification of the following two CBS requirements:

The following is a list of CBS requirements selected for verification during this activity

Table 6. CBS requirements

Number	Requirement	Verification Method
CBS-020	The CBS shall be capable of maintaining a continuous connection to a Network Transport of RTCM via Internet Protocol (NTRIP) caster for each of the known RBS locations	Demonstration
CBS-021	The CBS shall be capable of sending RTCM correction messages to the corresponding RBS within 2 seconds of reception of RTCM information from an NTRIP caster.	Analysis

No specific test cases were developed for the verification of the CBS requirements. The CBS was collecting data as the regression field test was being performed. This data was used for the verification of the CBS requirements. Figure 25 presents a subset of the data the CBS was collecting during the test event.

Message_ID	Topic	Subject	Content	Content	Received	Stored
a52f1ac5-1a44-4b59-9330-c4f2e5ef9af2	J2735.SPAT	14815 SPAT	00133100385A161E9D32ECD995057CBCFA209597366CBCB9EFD39DFC	17:30:00.077	17:30:00.110	
9aeeb83e-8b3d-4bf7-ae64-ae2d1a127023	J2735.SPAT	14815 SPAT	00133100385A161E9D32ECD995057CBCFA209597366CBCB9EFD39DFC	17:30:00.137	17:30:00.170	
fca2635b-ca98-4bc5-ab2f-55e2e450b2ec	J2735.SPAT	14815 SPAT	00133100385A161E9D32ECD995057CBCFA209597366CBCB9EFD39DFC	17:30:00.233	17:30:00.280	
3ca50ac0-63fd-4106-82d7-dda1e6aac23e	J2735.SPAT	14815 SPAT	00133100385A161E9D32ECD995057CBCFA209597366CBCB9EFD39DFC	17:30:00.340	17:30:00.387	
ad091850-63ad-449a-9b11-5f5088a16acd	J2735.MAP	14815 MapData	001280EC380530203073BE154D75DE4439AB3522198C02DC051F702B2I	17:30:00.397	17:30:00.440	
c0eb3c7f-75af-482e-86d1-35a78b0cd3cb	J2735.MAP	14815 MapData	001280EC380530203073BE154D75DE4439AB3522198C02DC051F702B2I	17:30:00.427	17:30:00.473	
f5779e5e-c629-4aea-9f39-52879fe3a2c1	J2735.SPAT	14815 SPAT	00133100385A161E9D32ECD995057CBCFA209597366CBCB9EFD39DFC	17:30:00.437	17:30:00.490	

Figure 25. RBS data received by the CBS

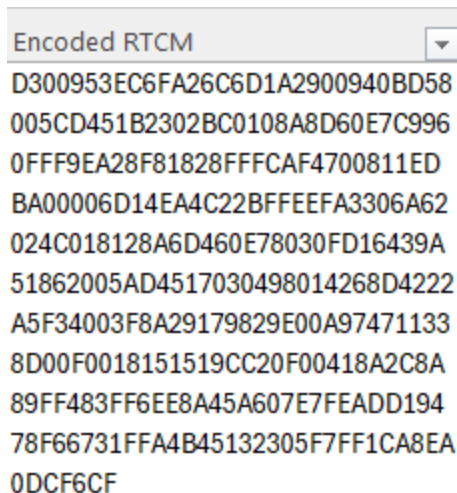
Requirement CBS-020 was verified via demonstration as the CBS was able to connect to the corresponding Network Transport of RTCM, send the data to the RBS and broadcast it via radio to the VBS. Allowing the VBS to achieve a Real-Time Kinematic (RTK) solution (See Reception and Application of RTCM corrections by the VBS). In addition we were able to match log files from the CBS down to the RSU showing the transmission of this message

An example can be seen below. The log files present the traceability of one RTCM message from the CBS down to the RSU. Figure 26 shows the reception by the CBS of an RTCM message

```
Encoded RTCM
d300953ec6fa26c6d1a2900940bd58005cd451b2
302bc0108a8d60e7c9960fff9ea28f81828ffcaf470
0811edba00006d14ea4c22bffeefa3306a62024c0
18128a6d460e78030fd16439a51862005ad451703
0498014268d4222a5f34003f8a29179829e00a974
711338d00f0018151519cc20f00418a2c8a89ff483f
f6ee8a45a607e7feadd19478f66731ffa4b4513230
5f7ff1ca8ea0dcf6cf
```

Figure 26. RTCM message received by the CBS from the ODOT .

Figure 27 shows the same message being received by the RBS. Unfortunately, the CBS was not programmed to log the time of arrival of the RTCM message .



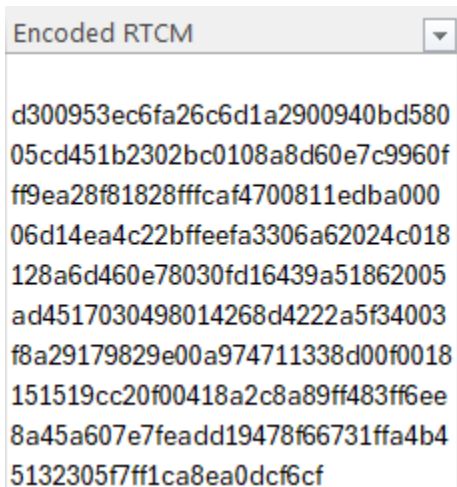
```

Encoded RTCM
D300953EC6FA26C6D1A2900940BD58
005CD451B2302BC0108A8D60E7C996
0FFF9EA28F81828FFFC4F4700811ED
BA00006D14EA4C22BFFEEFA3306A62
024C018128A6D460E78030FD16439A
51862005AD4517030498014268D4222
A5F34003F8A29179829E00A97471133
8D00F0018151519CC20F00418A2C8A
89FF483FF6EE8A45A607E7FEADD194
78F66731FFA4B45132305F7FF1CA8EA
0DCF6CF

```

Figure 27. RTCM message received by the RBS from the CBS.

And finally in Figure 28 we can track the same message as it was received by the RSU prior to being broadcasted to the VBS:



```

Encoded RTCM
d300953ec6fa26c6d1a2900940bd580
05cd451b2302bc0108a8d60e7c9960f
ff9ea28f81828fffc4f4700811edba000
06d14ea4c22bffeefa3306a62024c018
128a6d460e78030fd16439a51862005
ad4517030498014268d4222a5f34003
f8a29179829e00a974711338d00f0018
151519cc20f00418a2c8a89ff483ff6ee
8a45a607e7feadd19478f66731ffa4b4
5132305f7ff1ca8ea0dcf6cf

```

Figure 28 RTCM Message received by the RSU from the RBS

Due to the design of the CBS there is currently not a feasible way to verify requirement CBS-021 - *The CBS shall be capable of sending RTCM correction messages to the corresponding RBS within 2 seconds of reception of RTCM information from an NTRIP caster.*

Summary of RCVW Regression Test performance

Reception and Application of RTCM corrections by the VBS

The RCVW is capable of connecting via the CBS to any public Real Time Network in order to obtain GNSS signal corrections (with proper credentials). These corrections are pushed down to the RBS and broadcasted to the VBS which are then used to adjust the received satellite signals and improve the accuracy of the positioning information. Prior to the test event, the CBS was configured to connect to the Ohio DOT's Real Time Network which allows a connection to the Ohio DOT's Continuously Reference Station . This connection provided the CBS with real-

time GPS/GNSS signal corrections to be broadcasted to the corresponding RBS. The RBS received these continuous corrections, processed and broadcast them over the C-V2X radio for the test vehicle to use. Once the vehicle subsystem received this information it was able to obtain a Real-Time Kinematic (RTK) GNSS solution which is equivalent to a sub-decimeter accuracy in the vehicle location. Figure 29 shows a subset of the log files from the test event. The RTK field in the logs show “fixed”, indicating that the GNSS has achieved a fixed RTK solution.

```
"[2024-12-13-21:21:00.226]-WPlugin/src/RCVWPlugin.cpp(839)--DEBUG--:LOC-TIME,LOC-SPEED,LOC-HEADING,SPEED,HEADING,RTK,FREQUENCY:1734124859800,-0.006,-75.2515,-0.006,-75.2515,fixed,3.838"¶
"[2024-12-13-21:21:00.518]-WPlugin/src/RCVWPlugin.cpp(839)--DEBUG--:LOC-TIME,LOC-SPEED,LOC-HEADING,SPEED,HEADING,RTK,FREQUENCY:1734124860000,-0.004,-75.2514,-0.004,-75.2514,fixed,3.86901"¶
"[2024-12-13-21:21:00.800]-WPlugin/src/RCVWPlugin.cpp(839)--DEBUG--:LOC-TIME,LOC-SPEED,LOC-HEADING,SPEED,HEADING,RTK,FREQUENCY:1734124860300,-0.004,-75.2512,-0.004,-75.2512,fixed,3.84768"¶
"[2024-12-13-21:21:00.995]-WPlugin/src/RCVWPlugin.cpp(839)--DEBUG--:LOC-TIME,LOC-SPEED,LOC-HEADING,SPEED,HEADING,RTK,FREQUENCY:1734124860600,-0.004,-75.2512,-0.004,-75.2512,fixed,3.82732"¶
```

Figure 29. RTK – Fixed Solution

Several GNSS stability issues were encountered (See Figure 30). The RCVW system is configured to issue a fault when the following issues with the GNSS signal are detected:

- Loss of GNSS signal
- Loss of RTK fix
- Position data reception at a rate less than 10 hz

During testing, the VBS would issue a Loss of RTK fix fault message at random times. After an investigation it was discovered that the GNSS device would lose configuration during these situations and needed to be reprogrammed. The device is capable of storing configuration files even after a power loss. The source of the issue was not identified, and further investigation is required.

```
"[2024-12-13-20:58:03.602]-WPlugin/src/RCVWPlugin.cpp(839)--DEBUG--:LOC-TIME,LOC-SPEED,LOC-HEADING,SPEED,HEADING,RTK,FREQUENCY:1734123483100,-20.324,-244.374,-20.324,-244.374,none,3.40923"¶
"[2024-12-13-20:58:03.801]-WPlugin/src/RCVWPlugin.cpp(839)--DEBUG--:LOC-TIME,LOC-SPEED,LOC-HEADING,SPEED,HEADING,RTK,FREQUENCY:1734123483400,-20.011,-244.258,-20.011,-244.258,none,3.40656"¶
"[2024-12-13-20:58:04.003]-WPlugin/src/RCVWPlugin.cpp(839)--DEBUG--:LOC-TIME,LOC-SPEED,LOC-HEADING,SPEED,HEADING,RTK,FREQUENCY:1734123483600,-19.877,-244.457,-19.877,-244.457,none,3.44441"¶
"[2024-12-13-20:58:04.223]-WPlugin/src/RCVWPlugin.cpp(839)--DEBUG--:LOC-TIME,LOC-SPEED,LOC-HEADING,SPEED,HEADING,RTK,FREQUENCY:1734123483800,-19.731,-244.641,-19.731,-244.641,none,3.48176"¶
```

Figure 30. RTK – No RTK Solution

In HRI Distinct Alerts based on Status of the HRI.

This issue was identified during the first round of testing (see Vehicle in HRI Warning vs Inform Alerts). After an analysis was performed, it was determined that the issue was a result of a misconfiguration of the system and the absence of the correct image to be issued when a vehicle was in the HRI with a train approaching. During regression testing the same issue occurred where the RCVW system is not able to make a distinction whether a train is approaching versus a train not approaching when a vehicle is within the HRI. The system issues an alert for the driver to move the vehicle away from the HRI, but it is not capable of distinguishing whether a train is approaching or not.

Further investigation is required.

GNSS Stability

Test case 4.2 “GNSS Stability and Accuracy (Observation)” was designed to validate the stability of the GNSS system by confirming the following 1) that a System Available message was issued to the driver when the vehicle is positioned at the edge of the HRI and remains in the same state as the vehicle remains static in the location. The test started at 4:25:35 as can be seen in Figure 31 the system issued an RCVW warning as the vehicle approached the HRI. It then proceeded to clear the previously issued Inform and Warning messages and issued a HRI Warning indicating the vehicle is stopped within the HRI. See Figure 31

```
[2024-12-13 21:25:36.703] Plugin/src/RCVWPlugin.cpp (1213) - DEBUG : Sending Application Message: ApproachWarning"
[2024-12-13 21:25:41.556] Plugin/src/RCVWPlugin.cpp (1207) - DEBUG : Sending Application Message: Clear ApproachInform"
[2024-12-13 21:25:41.566] Plugin/src/RCVWPlugin.cpp (1219) - DEBUG : Sending Application Message: Clear ApproachWarning"
[2024-12-13 21:25:41.799] Plugin/src/RCVWPlugin.cpp (1225) - DEBUG : Sending Application Message: HRIWarning"
```

Figure 31. RCVW Warnings Near To and On-Track

The HRI Warning message remains until the system loses RTK data for around 1.5 seconds (we are not clear on why this occurred) and a fault message is issued. See Figure 32.

```
[2024-12-13 21:26:21.785] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: RTK Data Not Received"
[2024-12-13 21:26:21.794] Plugin/src/RCVWPlugin.cpp (1195) - DEBUG : Sending Application Message: Clear Available"
[2024-12-13 21:26:21.804] Plugin/src/RCVWPlugin.cpp (1234) - DEBUG : Sending Application Message: Clear HRIWarning"
```

Figure 32. RCVW Warnings Following RTK Loss

The RTK data is regained, and the system remains stable until the end of the test when the vehicle starts to move away from the HRI. See Figure 33

```
[2024-12-13 21:26:21.914] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
[2024-12-13 21:26:23.336] Plugin/src/RCVWPlugin.cpp (1189) - DEBUG : Sending Application Message: Available"
[2024-12-13 21:26:23.350] Plugin/src/RCVWPlugin.cpp (1225) - DEBUG : Sending Application Message: HRIWarning"
[2024-12-13 21:29:18.114] Plugin/src/RCVWPlugin.cpp (1234) - DEBUG : Sending Application Message: Clear HRIWarning"
```

Figure 33. RCVW Clear On-Track Warning

This data verifies that the stability of the GNSS system is up to what is required, as long as GNSS signal is available and RTK corrections are received.

Flickering Fault message

This issue was identified during the first round of testing (see Flickering Fault). The initial investigation pointed to a coding error. However, due to time constraints the software was not able to find a solution. The issue persisted during regression testing. Further investigation is required. See Figure 34

```
"[2024-12-13 21:31:19.103] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: MAP Data Not Received"
"[2024-12-13 21:31:19.214] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:31:40.637] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
"[2024-12-13 21:31:40.748] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:31:42.681] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
"[2024-12-13 21:31:42.792] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:31:44.725] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
"[2024-12-13 21:31:44.836] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:31:46.769] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
"[2024-12-13 21:31:46.879] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:31:48.818] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: MAP Data Not Received"
"[2024-12-13 21:31:48.928] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:31:50.862] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: MAP Data Not Received"
"[2024-12-13 21:31:50.973] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
```

Figure 34. Error Message Cleared by the System

Loss of Heartbeat

Test case 3.5 *Loss of IEEE 1570 Heartbeat* was designed to verify that a Loss of Heartbeat fault message is issued when the VBS detected a loss of heartbeat from a connected IEEE-1570 compliant device. The test case required a disconnection of the IEEE-1570 device from the RBS at some point during a vehicle approach and observe the issued fault message. During the execution of the test case, the IEEE-1570 was disconnected and instead of a Loss of Heartbeat fault message, a Loss of SPAT fault message was issued by the system (Figure 35). In addition, when the IEEE-1570 device was reconnected to the RBS, the system failed to reinitialize the corresponding plug in resulting and the issued fault not being able to clear. Further investigation to diagnose the issue is required.

```
"[2024-12-13 21:06:11.181] Plugin/src/RCVWPlugin.cpp (1201) - DEBUG : Sending Application Message: ApproachInform"
"[2024-12-13 21:06:14.905] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
"[2024-12-13 21:06:14.915] Plugin/src/RCVWPlugin.cpp (1195) - DEBUG : Sending Application Message: Clear Available"
"[2024-12-13 21:06:14.926] Plugin/src/RCVWPlugin.cpp (1207) - DEBUG : Sending Application Message: Clear ApproachInform"
"[2024-12-13 21:06:15.037] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:06:16.970] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
"[2024-12-13 21:06:17.081] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:06:19.015] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
"[2024-12-13 21:06:19.125] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:06:21.059] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
"[2024-12-13 21:06:21.169] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:06:23.103] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
"[2024-12-13 21:06:23.213] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:06:25.148] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
"[2024-12-13 21:06:25.258] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:06:27.193] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
"[2024-12-13 21:06:27.305] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:06:29.238] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
"[2024-12-13 21:06:29.349] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:06:31.283] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
"[2024-12-13 21:06:31.393] Plugin/src/RCVWPlugin.cpp (1246) - DEBUG : Sending Application Message: Clear Error"
"[2024-12-13 21:06:33.337] Plugin/src/RCVWPlugin.cpp (1240) - DEBUG : Sending Application Message: Error: SPAT Data Not Received"
```

Figure 35. Log File of the Loss of IEEE 1570 Test Case

Detailed RCVW Field Test & Evaluation Results

The objective of the verification testing comprising the RCVW FT&E Plan was to verify and demonstrate that the RCVW system possesses the functionality and performance necessary to deliver the functions and benefits proposed for the system described in the RCVW Concept of Operations [1] in an operational setting. FT&E testing also aimed to verify that the RCVW system meets the requirements documented in the System Requirements Specification [2]. The following pages of this section detail the results of each test case and includes noted observations.

Results for the Regression Testing can be found in section Regression Testing Results

Test Case No. and Title	1.1. CV Approaches and Travels through HRI with no Train Approaching, 35 mph – No HRI Active Signal
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: - Confirm that an RCVW is NOT displayed/annunciated while the CV is within the Approach and HRI zones - Confirm that a System Ready is displayed when the CV is outside the Approach Zone and system functioning - Confirm that a System Available is displayed as the CV enters the approach zone and the preemption signal is not active - Confirm that a System Available is displayed while the CV is entering or departing from the HRI - Confirm that a System Ready is displayed once the CV has departed the HRI Secondary Objectives: - Confirm RBS broadcasts the MAP message at a rate of 1Hz - Confirm RBS broadcasts the SPaT message at a rate of 10Hz with the appropriate approach lane status when the HRI is not active
Requirements Verified	RCVW-007 VBS-,001, 004, 006, 007a, 007b,008, 010, 014,022,023, 025 RBS-007, 007b, 007c, 019, 022, 022a
Brief Description	The primary purpose of this test is to verify that a RCVW is not generated because a train has not been detected (i.e., a preempt call has not been placed) by the highway rail grade crossing train detection system. 3 iterations will be conducted following the same conditions described in this test case to collect multiple sets of data for later analysis.
Test Setup and Configuration	Prior to beginning the test, the Test Administrator shall verify that the following conditions are planned for this test: - HRI status Not Active for the duration of this test - The initial approach speed of the CV shall be 35 mph

Test Case No. and Title	1.1. CV Approaches and Travels through HRI with no Train Approaching, 35 mph – No HRI Active Signal	
	- Pavement surface condition is dry. - The status of the RCVW system is “Normal” - This test will be performed using a passenger car (type A). The following configuration states should be verified prior to test execution: - Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X radios) - Ensure the RBS is powered and operating normally (MAP and SPaT are broadcast) Logging is activated	
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed: - the algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. - the algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the test director and algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets	
Test Case No. and Title	1.1. CV Approaches and Travels through HRI with no Train Approaching, 35 mph – No HRI Active Signal	
Verification Phase	II – Field Test Verification	
Pass/Fail	☐ Pass (met all expected results) ☒ Fail (did not meet one or more expected results)	Date: 7/25/2024

Test Case No. and Title	1.1. CV Approaches and Travels through HRI with no Train Approaching, 35 mph – No HRI Active Signal		
Verification Phase	II – Field Test Verification		
Expected Results [requirement]	Met?		Notes
	Y	N	
An RCVW is not displayed/annunciated while the CV is within the approach and no preemption is detected [RCVW-007] [VBS-006] [VBS-007a] [VBS-007b] [VBS-10] [VBS-014] [VBS-025] [RBS-007c]	Y		
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
A System Available is displayed while the CV is moving across the HRI [RCVW-007] [VBS-001] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
A System Available is displayed when the CV enters the approach area and the preemption signal is not active [RCVW-007] [VBS-001] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
RBS broadcasts the HRI Geometry once per second and indicate the HRI ID [RBS-007] [RBS-007a] [RBS-007b]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]		N	The averages are 9.55 messages per second.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS- 004a] [RBS-004b]	Y		
The VBS shall be capable of receiving messages sent by the RBS within 50 ms.[VBS-022]		N	See RBS/VBS Time Discrepancies
The VBS shall be capable of processing received data within 85 ms.[VBS-023]	Y		

Test Case No. and Title		1.1. CV Approaches and Travels through HRI with no Train Approaching, 35 mph – No HRI Active Signal	
Verification Phase		II – Field Test Verification	
Expected Results [requirement]	Met?		Notes
	Y	N	
The RBS shall be capable of generating and broadcasting RTK corrections using the RTCM messaging protocol. [RBS-022] [RBS-022a]		N	See Verification of Requirement RBS-022.
All log files shall be time stamped. [RBS-019]	Y		
Notes			

Test Case No and Title	1.2 Type A CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: • Confirm that the System Ready graphic is displayed when the CV is outside the Approach Zone and system functioning • Confirm that the active preemption can be detected by the RBS by using a voltage-based connection and the IEEE-1570 interface • Confirm that an Approach Inform Alert is displayed/annunciated while the CV is within the approach zone and approaching the HRI, there is an active Preemption signal and an RCVW is not predicted • Confirm that an RCVW is displayed/annunciated while the CV is within the Approach zone and approaching the HRI, there is an active Preemption Signal, and a RCVW is predicted • Confirm that an RCVW is no longer displayed/annunciated while the CV is within the Approach zone and approaching the HRI as a result of Driver Action (CV decelerates), there is an active Preemption Signal, and a RCVW is not/no longer predicted • Confirm that an Approach Inform Alert will be reissued while the CV is in the Approach Zone, preemption is active and RCVW is not predicted • Confirm that the Approach Inform Alert is extinguished after preemption deactivates. • Confirm that a System Active graphic is displayed while the CV is moving across the HRI (if applicable) • Confirm that a System Ready is displayed once the CV has departed the HRI (if applicable) • Analyze log files to confirm that warnings and alerts are accurate and correspond to displayed results Secondary Objectives: • Confirm RBS broadcasts the MAP message at a rate of 1Hz • Confirm RBS broadcasts the SPaT message at a rate of 10Hz with the appropriate approach lane status when the HRI is active •
Requirements Verified	RCVW-007 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 008, 010, 011b, 011c, 012, 013, 014, 016, 017, 020, 021, 022, 023, 025

Test Case No and Title	1.2 Type A CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
	RBS-002b, 004, 004a, 004b, 007, 007a, 007b, 007c, 019, 022, 022a
Brief Description	The primary purpose of this test is to verify that an RCVW is generated with the CV within the Approach zone and approaching the HRI, an active Preemption Signal, and a RCVW predicted for the specified conditions. Additionally, the test is to verify that the RCVW is no longer generated once the driver takes satisfactory action (CV decelerates) to permit safe stoppage prior to the HRI. This test case will verify that the RBS is capable of detecting the preemption signal by using a voltage-based connection and the IEEE-1570 interface. Three iterations using the voltage-based connection and 3 iterations using the IEEE-1570 device will be conducted following the same conditions described in this test case to collect multiple sets of data for later analysis.
Test Setup and Configuration	Prior to beginning the test, the Test director shall verify that the following conditions are planned for this test: • HRI status Active for the duration of this test • Voltage-based preemption and IEEE-1570 device are ready • The initial approach speed of the CV shall be 35 mph • Pavement surface condition is dry. • The status of the RCVW system is “Normal” • This test will be performed using a passenger car (type A). The following configuration states should be verified prior to test execution: • Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) • Ensure the RBS is powered and operating normally (MAP and SPaT are broadcast) • Logging is activated
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed: • The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case.

Test Case No and Title	1.2 Type A CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
	The algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case		
Pass/Fail	☐ Pass (met all expected results) ☒ Fail (did not meet one or more expected results)		Date: 7/25/2024
Test Case No and Title	1.2 Type A CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verifications		
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
A System Available is displayed while the CV is moving across the HRI [RCVW-007] [VBS-001] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]	Y		
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-011d] [VBS-012a] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]	Y		

Test Case No and Title	1.2 Type A CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
The RCVW will extinguish after proper driver reaction and an Approach Inform Alert will be reissued while the CV is in the Approach Zone, preemption is active and no RCVW is not predicted [VBS-011b]	Y		
RBS broadcasts the HRI Geometry once per second and indicates the HRI ID [RBS-007] [RBS-007a] [RBS-007b]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]		N	The averages are 9.55 messages per second.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS- 004a] [RBS-004b]	Y		
The VBS shall be capable of receiving messages sent by the RBS within 50 ms.[VBS-022]		N	See RBS/VBS Time Discrepancies
The VBS shall be capable of processing received data within 85 ms.[VBS-023]	Y		
The RBS shall be capable of determining HRI activation status using the DC voltage-based preemption signal from the HRI Controller and, indirectly, from an IEEE 1570-compliant serial device that has been interfaced to the HRI Controller, when present. [RBS-002a]	Y		Voltage-based activation was not tested during this event.
The RBS shall be capable of generating and broadcasting RTK corrections using the RTCM messaging protocol. [RBS-022] [RBS-022a]		N	See Verification of Requirement RBS-022.
All log files shall be time stamped. [RBS-019]	Y		
Notes			

Test Case No. and Title	1.3 Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: • Confirm that the System Ready graphic is displayed when the CV is outside the Approach Zone and system functioning • Confirm that a System Available is displayed as the CV enters the approach zone and the preemption signal is not active • Confirm that an Approach Inform Alert is displayed/annunciated while the CV is within the approach zone and approaching the HRI, there is an active Preemption signal and an RCVW is not predicted • Confirm that an RCVW is displayed/annunciated while the CV is within the Approach zone and approaching the HRI, there is an active Preemption Signal, and a RCVW is predicted • Confirm that an RCVW is no longer displayed/annunciated while the CV is within the Approach zone and approaching the HRI as a result of Driver Action (CV decelerates), there is an active Preemption Signal, and a RCVW is not/no longer predicted • Confirm that an Approach Inform Alert will be reissued while the CV is in the Approach Zone, preemption is active and RCVW is not predicted • Confirm that the Approach Inform Alert is extinguished after preemption deactivates. • Confirm that a System Available is displayed while the CV is moving across the HRI (if applicable) • Confirm that a System Ready is displayed once the CV has departed the HRI (if applicable) • Analyze log files to confirm that warnings and alerts are accurate and correspond to displayed results Secondary Objectives: • Compare results with those of test case 1.2 to gain an understanding of the RCVW safety envelope as impacted by varying the CV distance from the HRI when the Preemption Signal is detected • Confirm RBS broadcasts the MAP message at a rate of 1Hz • Confirm RBS broadcasts the SPaT message at a rate of 10Hz with the appropriate approach lane status when the HRI is active.
Requirements Verified	RCVW-007

Test Case No. and Title	1.3 Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
	VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 008, 010, 011b, 011c, 012, 013, 014, 016, 017, 020, 021, 022, 023, 025 RBS-004, 004a, 004b, 007, 007a, 007b, 007c, 019, 022, 022a
Brief Description	The primary purpose of this test is to verify that an RCVW is generated with the CV within the Approach zone and approaching the HRI, an active Preemption Signal, and a RCVW predicted for the specified conditions. Additionally, the test is to verify that the RCVW is no longer generated once the driver takes satisfactory action to permit safe stoppage prior to the HRI. This Test Case is similar to Test Case 1.2 except that the preemption detection becomes active when the CV is within the Approach Zone at a point $\frac{3}{4}$ the distance of the zone (based on initial approach entry speed), rather than at entry into the Approach Zone. 3 iterations will be conducted following the same conditions described in this test case to collect multiple sets of data for later analysis.
Test Setup and Configuration	Prior to beginning the test, the Test director shall verify that the following conditions are planned for this test: • HRI status Active when the CV reaches the 75% Approach Zone point • The initial approach speed of the CV shall be 35 mph • Pavement surface condition is dry. • The status of the RCVW system is “Normal” • This test will be performed using a passenger car (type A). The following configuration states should be verified prior to test execution: • Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) • Ensure the RBS is powered and operating normally (MAP and SPaT are broadcast) • Logging is activated
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed: • Coordination between the driver and test director via radio will take place to coordinate when the CV approaches the $\frac{3}{4}$ distance so the preemption can be activated.

Test Case No. and Title	1.3 Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
	- The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. - The algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets. Results will be compared with test case 1.2 to contrast the impact of the distance from HRI when Preemption Signal is detected		
Pass / Fail	☐ Pass (met all expected results) ☒ Fail (did not meet one or more expected results)		Date: 7/26/2024
Test Case No. and Title	1.3 Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
A System Available is displayed while the CV is moving across the HRI [RCVW-007] [VBS-001] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle	Y		

Test Case No. and Title	1.3 Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
speed and distance do not warrant an approach warning. [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]			
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-011d] [VBS-012a] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]	Y		
The RCVW will extinguish after proper driver reaction and an Approach Inform Alert will be reissued while the CV is in the Approach Zone, preemption is active and no RCVW is not predicted [VBS-011b]	Y		
RBS broadcasts the HRI Geometry once per second and indicate the HRI ID [RBS-007] [RBS-007a] [RBS-007b]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]		N	The averages are 9.55 messages per second.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS-004a] [RBS-004b]	Y		

Test Case No. and Title	1.3 Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
The VBS shall be capable of receiving messages sent by the RBS within 50 ms.[VBS-022]		N	See RBS/VBS Time Discrepancies
The VBS shall be capable of processing received data within 85 ms.[VBS-023]	Y		
The RBS shall be capable of generating and broadcasting RTK corrections using the RTCM messaging protocol. [RBS-022] [RBS-022a]		N	See Verification of Requirement RBS-022.
All log files shall be time stamped. [6]	Y		
Notes			

Test Case No. and Title	1.4 Type A CV Approaches with Preemption Signal Detected at the Approach Midpoint, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: • Confirm that the System Ready graphic is displayed when the CV is outside the Approach Zone and system functioning • Confirm that a System Available is displayed as the CV enters the approach zone and the preemption signal is not active • Confirm that an Approach Inform Alert is displayed/annunciated while the CV is within the approach zone and approaching the HRI, there is an active Preemption signal and an RCVW is not predicted • Confirm that an RCVW is displayed/annunciated while the CV is within the Approach zone and approaching the HRI, there is an active Preemption Signal, and a RCVW is predicted • Confirm that an RCVW is no longer displayed/annunciated while the CV is within the Approach zone and approaching the HRI as a result of Driver Action (CV decelerates), there is an active Preemption Signal, and a RCVW is not/no longer predicted • Confirm that an Approach Inform Alert will be reissued while the CV is in the Approach Zone, preemption is active and RCVW is not predicted • Confirm that the Approach Inform Alert is extinguished after preemption deactivates. • Analyze log files to confirm that warnings and alerts are accurate and correspond to displayed results Secondary Objectives: • Compare results with those of test case 1.2 to gain an understanding of the RCVW safety envelope as impacted by varying the CV distance from the HRI when the Preemption Signal is detected • Confirm RBS broadcasts the MAP message at a rate of 1Hz • Confirm RBS broadcasts the SPaT message at a rate of 10Hz with the appropriate approach lane status when the HRI is active
Requirements Verified	RCVW-007 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 008, 010, 011b, 011c, 012, 013, 014, 016, 017, 020, 021, 022, 023, 025 RBS-004, 004a, 004b, 007, 007a, 007b, 007c, 019, 022, 022a

Test Case No. and Title	1.4 Type A CV Approaches with Preemption Signal Detected at the Approach Midpoint, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
Brief Description	The primary purpose of this test is to verify that an RCVW is generated with the CV within the Approach zone and approaching the HRI, an active Preemption Signal, and a RCVW predicted for the specified conditions. Additionally, the test is to verify that the RCVW is no longer generated once the driver takes satisfactory action to permit safe stoppage prior to the HRI. This Test Case is similar to Test Case 1.2, except that the Preemption Signal detection becomes active when the CV is within the Approach Zone at a point 1/2 the distance of the Approach zone (based on initial approach entry speed), rather than at entry into the Approach Zone. 3 iterations will be conducted following the same conditions described in this test case to collect multiple sets of data for later analysis.
Test Setup and Configuration	Prior to beginning the test, the Test director shall verify that the following conditions are planned for this test: • HRI status Active when the CV reaches the 50% Approach Zone point • The initial approach speed of the CV shall be 35 mph • Pavement surface condition is dry. • The status of the RCVW system is “Normal” • This test will be performed using a passenger car (type A). The following configuration states should be verified prior to test execution: • Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) • Ensure the RBS is powered and operating normally (MAP and SPaT are broadcast) • Logging is activated

Test Case No. and Title	1.4 Type A CV Approaches with Preemption Signal Detected at the Approach Midpoint, 35 mph, Dry Conditions	
Verification Phase	II – Field Test Verification	
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed: • Coordination between the driver and test director via radio will take place to coordinate when the CV approaches the ½ distance so the preemption can be activated. • The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. • The algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets. Results will be compared with test case 1.2 to contrast the impact of the distance from HRI when Preemption Signal is detected	
Pass / Fail	☐ Pass (met all expected results) ☒ Fail (did not meet one or more expected results)	Date: 7/26/2024

Test Case No. and Title			1.4 Type A CV Approaches with Preemption Signal Detected at the Approach Midpoint, 35 mph, Dry Conditions	
Verification Phase			II – Field Test Verification	
Expected Results [requirement]	Met?		Notes	
	Y	N		
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y			

Test Case No. and Title			1.4 Type A CV Approaches with Preemption Signal Detected at the Approach Midpoint, 35 mph, Dry Conditions
Verification Phase			II – Field Test Verification
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Available is displayed while the CV is moving across the HRI [RCVW-007] [VBS-001] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]	Y		
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-011d] [VBS-012a] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]	Y		
The RCVW will extinguish after proper driver reaction and an Approach Inform Alert will be reissued while the CV is in the Approach Zone, preemption is active and no RCVW is not predicted [VBS-011b]	Y		
RBS broadcasts the HRI Geometry once per second and indicate the HRI ID [RBS-007] [RBS-007a] [RBS-007b]	Y		

Test Case No. and Title			1.4 Type A CV Approaches with Preemption Signal Detected at the Approach Midpoint, 35 mph, Dry Conditions
Verification Phase			II – Field Test Verification
Expected Results [requirement]	Met?		Notes
	Y	N	
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]		N	The averages are 9.55 messages per second.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS- 004a] [RBS-004b]	Y		
The VBS shall be capable of receiving messages sent by the RBS within 50 ms.[VBS-022]		N	See RBS/VBS Time Discrepancies
The VBS shall be capable of processing received data within 85 ms.[VBS-023]	Y		
The RBS shall be capable of generating and broadcasting RTK corrections using the RTCM messaging protocol. [RBS-022] [RBS-022a]		N	See Verification of Requirement RBS-022.
All log files shall be time stamped. [RBS-019]	Y		
Notes			

Test Case No. and Title	1.5 Type A CV Approaches with Preemption Signal Detected near the Approach Zone Hazard Zone Transition, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: • Confirm that the System Ready graphic is displayed when the CV is outside the Approach Zone and system functioning • Confirm that a System Available is displayed as the CV enters the approach zone and the preemption signal is not active • Confirm that an Approach Inform Alert is displayed/annunciated while the CV is within the approach zone and approaching the HRI, there is an active Preemption signal and an RCVW is not predicted • Confirm that an RCVW is displayed/annunciated while the CV is within the Approach zone and approaching the HRI, there is an active Preemption Signal, and a RCVW is predicted • Confirm that an RCVW is no longer displayed/annunciated while the CV is within the Approach zone and approaching the HRI as a result of Driver Action (CV decelerates), there is an active Preemption Signal, and a RCVW is not/no longer predicted • Confirm that an Approach Inform Alert will be reissued while the CV is in the Approach Zone, preemption is active and RCVW is not predicted • Confirm that the Approach Inform Alert is extinguished after preemption deactivates. • Confirm that a System Available is displayed while the CV is moving across the HRI (if applicable) • Confirm that a System Ready is displayed once the CV has departed the HRI (if applicable) • Analyze log files to confirm that warnings and alerts are accurate and correspond to displayed results Secondary Objectives: • Compare results with those of test case 1.2 to gain an understanding of the RCVW safety envelope as impacted by varying the CV distance from the HRI when the Preemption Signal is detected • Confirm RBS broadcasts the MAP message at a rate of 1Hz • Confirm RBS broadcasts the SPaT message at a rate of 10Hz with the appropriate approach lane status when the HRI is active.

Test Case No. and Title	1.5 Type A CV Approaches with Preemption Signal Detected near the Approach Zone Hazard Zone Transition, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
Requirements Verified	RCVW-007 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 008, 010, 011b, 011c, 012, 013, 014, 016, 017, 020, 021, 022, 023, 025 RBS-004, 004a, 004b, 007, 007a, 007b, 007c, 019, 022, 022a
Brief Description	The primary purpose of this test is to verify that an RCVW is generated with the CV within the Approach zone and approaching the HRI, an active Preemption Signal, and a RCVW predicted for the specified conditions. Additionally, the test is to verify that the RCVW is no longer generated once the driver takes satisfactory action to permit safe stoppage prior to the HRI. This Test Case is similar to Test Case 1.2, except that the preemption detection becomes active when the CV is within the Approach Zone at a point near the transition from the Approach Zone to the HRI Zone (approximately 90% of the distance of the approach zone based on initial approach entry speed), rather than at entry into the Approach Zone. 3 iterations will be conducted following the same conditions described in this test case to collect multiple sets of data for later analysis.
Test Setup and Configuration	Prior to beginning the test, the Test Administrator shall verify that the following conditions are planned for this test: • HRI status Active when the CV reaches the 90% Approach Zone point • The initial approach speed of the CV shall be 35 mph • Pavement surface condition is dry. • The status of the RCVW system is “Normal” • This test will be performed using a passenger car (type A). The following configuration states should be verified prior to test execution: • Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2x) • Ensure the RBS is powered and operating normally (MAP and SPaT are broadcast) • Logging is activated

Test Case No. and Title	1.5 Type A CV Approaches with Preemption Signal Detected near the Approach Zone Hazard Zone Transition, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed: • Coordination between the driver and test director via radio will take place to coordinate when the CV approaches the 90% distance so the preemption can be activated. • The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. • The algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets. Results will be compared with test case 1.2 to contrast the impact of the distance from HRI when Preemption Signal is detected		
Pass / Fail	☐ Pass (met all expected results) ☒ Fail (did not meet one or more expected results)		Date: 7/26/2024
Test Case No. and Title		1.5 Type A CV Approaches with Preemption Signal Detected near the Approach Zone Hazard Zone Transition, 35 mph, Dry Conditions	
Verification Phase		II – Field Test Verification	
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b]	Y		

Test Case No. and Title	1.5 Type A CV Approaches with Preemption Signal Detected near the Approach Zone Hazard Zone Transition, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
[VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]			
A System Available is displayed while the CV is moving across the HRI [RCVW-007] [VBS-001] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]	Y		
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-011d] [VBS-012a] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]	Y		

Test Case No. and Title	1.5 Type A CV Approaches with Preemption Signal Detected near the Approach Zone Hazard Zone Transition, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
The RCVW will extinguish after proper driver reaction and an Approach Inform Alert will be reissued while the CV is in the Approach Zone, preemption is active and no RCVW is not predicted [VBS-011b]	Y		
RBS broadcasts the HRI Geometry once per second and indicate the HRI ID [RBS-007] [RBS-007a] [RBS-007b]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]		N	The averages are 9.55 messages per second.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS- 004a] [RBS-004b]	Y		
The VBS shall be capable of receiving messages sent by the RBS within 50 ms.[VBS-022]		N	See RBS/VBS Time Discrepancies
The VBS shall be capable of processing received data within 85 ms.[VBS-023]	Y		
The RBS shall be capable of generating and broadcasting RTK corrections using the RTCM messaging protocol. [RBS-022] [RBS-022a]		N	See Verification of Requirement RBS-022.
All log files shall be time stamped. [RBS-019]	Y		
Notes			

Test Case No. and Title	1.6. Type A CV Approaches with Preemption Signal Detected at Approach Entry, Preemption signal is removed at Midpoint, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: • Confirm that a System Ready is displayed when the CV is outside the Approach Zone and system functioning • Confirm that an Approach Inform Alert is displayed/annunciated while the CV is within the approach zone and approaching the HRI, there is an active Preemption signal and an RCVW is not predicted • Confirm that an RCVW is displayed/annunciated while the CV is within the Approach zone and approaching the HRI, there is an active Preemption Signal, and a RCVW is predicted • Confirm that an RCVW is no longer displayed/annunciated while the CV is within the Approach zone and approaching the HRI as a result of the Preemption signal removed • Confirm that an Informed Alert reappears while the CV remains in the Approach Zone and approaching the HRI, there is an active Preemption signal and an RCVW is not predicted • Confirm that the Approach Inform Alert is extinguished after preemption deactivates. • Confirm that a System Available is displayed while the CV is moving across the HRI (if applicable) • Confirm that a System Ready is displayed once the CV has departed the HRI (if applicable) Secondary Objectives: • Confirm RBS broadcasts the MAP message at a rate of 1Hz • Confirm RBS broadcasts the SPaT message at a rate of 10Hz with the appropriate approach lane status when the HRI is active • Confirm RBS broadcasts the SPaT message at a rate of 10Hz with the appropriate approach lane status when the HRI is not active •
Requirements Verified	RCVW-007 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 008, 010, 011b, 011c, 012, 013, 014, 016, 017, 020, 021, 022, 023, 025 RBS-004, 004a, 004b, 007, 007a, 007b, 007c, 019, 022, 022a

Test Case No. and Title	1.6. Type A CV Approaches with Preemption Signal Detected at Approach Entry, Preemption signal is removed at Midpoint, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
Brief Description	The primary purpose of this test is to verify that an RCVW is generated with the CV within the Approach zone and approaching the HRI, an active Preemption Signal, and a RCVW predicted for the specified conditions. Additionally, the test is to verify that the RCVW is no longer issued once the Preemption signal is no longer present. 3 iterations will be conducted following the same conditions described in this test case to collect multiple sets of data for later analysis.
Test Setup and Configuration	Prior to beginning the test, the Test director shall verify that the following conditions are planned for this test: • HRI Status active at/prior to CV Approach Zone entry • The initial approach speed of the CV shall be 35 mph • Pavement surface condition is dry. • The status of the RCVW system is “Normal” • At the Approach Zone midpoint, Preemption signal will be turned off • This test will be performed using a passenger car (type A). • No driver response, the vehicle is to advance through the HRI. The following configuration states should be verified prior to test execution: • Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) • Ensure the RBS is powered and operating normally (MAP and SPaT are broadcast) • Logging is activated
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed: • Coordination between the driver and test director via radio will take place to coordinate when the CV approaches the ½ distance so the preemption can be deactivated.

Test Case No. and Title	1.6. Type A CV Approaches with Preemption Signal Detected at Approach Entry, Preemption signal is removed at Midpoint, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
	- The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. - the algorithm shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets		
Pass / Fail	☐ Pass (met all expected results) ☒ Fail (did not meet one or more expected results)		Date: 7/26/2024
Test Case No. and Title		1.6. Type A CV Approaches with Preemption Signal Detected at Approach Entry, Preemption signal is removed at Midpoint, 35 mph, Dry Conditions	
Verification Phase		II – Field Test Verification	
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004]	Y		

Test Case No. and Title	1.6. Type A CV Approaches with Preemption Signal Detected at Approach Entry, Preemption signal is removed at Midpoint, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
[VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]			
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]	Y		
The RCVW extinguishes as preemption signal is removed [VBS-011b]	Y		
RBS broadcasts the HRI Geometry once per second and indicate the HRI ID [RBS-007] [RBS-007a] [RBS-007b]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]		N	The averages are 9.55 messages per second.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS- 004a] [RBS-004b]	Y		
The VBS shall be capable of receiving messages sent by the RBS within 50 ms.[VBS-022]		N	See RBS/VBS Time Discrepancies

Test Case No. and Title	1.6. Type A CV Approaches with Preemption Signal Detected at Approach Entry, Preemption signal is removed at Midpoint, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
The VBS shall be capable of processing received data within 85 ms.[VBS-023]	Y		
The RBS shall be capable of generating and broadcasting RTK corrections using the RTCM messaging protocol. [RBS-022] [RBS-022a]		N	See Verification of Requirement RBS-022.
All log files shall be time stamped. [RBS-019]	Y		
Notes			

Test Case No. and Title	2.1. CV Stopped in HRI Hazard Zone, Preemption not active/active
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: • Confirm that a System Ready is displayed when the CV is outside the Approach Zone and system functioning • Confirm that a system available is displayed when the CV is in the Approach Zone • Confirm that a Unique Warn alert to clear the HRI is displayed when the CV is stopped in the HRI and the preemption is active. • Confirm that a Unique alert to clear the HRI is displayed when the CV is stopped in the HRI and the preemption is not active. • Confirm that a System Ready is displayed when the CV leaves the HRI Secondary Objectives: • Confirm RBS broadcasts the SPaT message at a rate of 10Hz with the appropriate approach lane status when the HRI is not active • Confirm RBS broadcasts the SPaT message at a rate of 10Hz with the appropriate approach lane status when the HRI is active
Requirements Verified	RCVW-007 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 008, 010, 011b, 011c, 011d, 012a, 012, 013, 014, 016, 017, 020, 021, 022, 023, 025, 026 RBS-004, 004a, 004b, 007, 007a, 007b, 007c, 019, 022, 022a
Brief Description	In this test case, the CV will approach the grade crossing, and then stop on the tracks (in the HRI Hazard Zone). A Unique Warn alert to clear the HRI shall be issued for the vehicle stopped in the HRI Hazard zone once the preemption is switched on. The warning will be removed once preemption is turned off, and replaced with a unique Alert to vacate the HRI. All messages are removed once the CV departs the HRI Hazard Zone. 3 iterations of each case will be conducted following the same conditions described in this test case to collect multiple sets of data for later analysis
Test Setup and Configuration	Prior to beginning the test, the Test director shall verify that the following conditions are planned for this test: • The CV approaches the HRI and stops on the tracks in the HRI Hazard Zone • With preemption off, an alert will be issued to vacate the HRI

Test Case No. and Title	2.1. CV Stopped in HRI Hazard Zone, Preemption not active/active		
Verification Phase	II – Field Test Verification		
	• With preemption on, a warning will be issued to vacate the HRI • The CV will depart the HRI after the alert and warning have been issued • The status of the RCVW system is “Normal” • This test will be performed using a passenger car (type A). The following configuration states should be verified prior to test execution: • Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) • Ensure the RBS is powered and operating normally (MAP and SPaT are broadcast) • Logging is activated		
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5. Error! Reference source not found. above. After the test, the following shall be performed: • The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. • the algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets.		
Pass / Fail	☐ Pass (met all expected results) ☒ Fail (did not meet one or more expected results)		Date: 7/26/2024
Test Case No. and Title		2.1. CV Stopped in HRI Hazard Zone, Preemption not active/active	
Verification Phase		II – Field Test Verification	
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a]	Y		

Test Case No. and Title	2.1. CV Stopped in HRI Hazard Zone, Preemption not active/active		
Verification Phase	II – Field Test Verification		
[VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]			
A System Available is displayed while the CV is moving across the HRI [RCVW-007] [VBS-001] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
The VBS shall continually issue unique HRI Hazard informs to the vehicle operator while the HRI safety devices are not active and the vehicle is stopped within the HRI Hazard Zone [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-011d] [VBS-012a] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025] [VBS-012a]	Y		
The VBS shall continually issue unique HRI Hazard warnings to the vehicle operator while the HRI safety devices are active and the vehicle is stopped within the HRI Hazard Zone. [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-011d] [VBS-012a] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025] [VBS-11d]	Y		
Confirm that the CV will not receive an Alert Warning it of clearing the HRI as its moving through it	Y		
Confirm that the System Available message remains displayed while any portion of the vehicle occupies the HRI and the System	Y		

Test Case No. and Title	2.1. CV Stopped in HRI Hazard Zone, Preemption not active/active		
Verification Phase	II – Field Test Verification		
Ready message displays only when all portions of the vehicle are clear of the HRI Approach Zone and HRI Hazard Zone [VBS-026]			
RBS broadcasts the HRI Geometry once per second and indicate the HRI ID [RBS-007] [RBS-007a] [RBS-007b]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]		N	The averages are 9.55 messages per second.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS-004a] [RBS-004b]	Y		
The VBS shall be capable of receiving messages sent by the RBS within 50 ms.[VBS-022]		N	See RBS/VBS Time Discrepancies
The VBS shall be capable of processing received data within 85 ms.[VBS-023]	Y		
The RBS shall be capable of generating and broadcasting RTK corrections using the RTCM messaging protocol. [RBS-022] [RBS-022a]		N	See Verification of Requirement RBS-022.
All log files shall be time stamped. [RBS-019]	Y		
Notes			

Test Case No. and Title	3.1 Loss of C-V2X after CV has received an RCVW
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: To demonstrate system's capability to detection and response to a loss of communications due to a failure with the C-V2X radio.
Requirements Verified	RCVW-007, VBS-001, 002, 003a, 003b, 003c, 006, 007, 007a, 007b, 008, 010, 011b, 011c, 012, 013014, 016, 017,018e, 018d, 019, 020, 025 RBS-005, 007c
Brief Description	The purpose of this test is to demonstrate how the RCVW system will behave when there is a loss of C-V2X after the system has issued an RCVW. As the CV enter the approach zone, with an active preemption signal, an approach inform will be displayed/annunciated. An RCVW will then be displayed/annunciated. As the driver begins to respond, there is a loss of C-V2X signal. The system will generate a graphic and audio message informing the driver of a System fault.
Test Setup and Configuration	Prior to beginning the test, the Test Director shall verify that the following conditions are planned for this test: - Train preemption signal detected at/prior to CV Approach Zone entry - The initial approach speed of the CV shall be 35 mph - The status of the RCVW system is "Normal" - This test will be performed using a CV passenger car (type A). - Driver response will be executed upon issuance of the RCVW, but prior to driver action resolving the RCVW C-V2X communications are lost. The following configuration states should be verified prior to test execution: - Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) - Logging is activated

Test Case No. and Title	3.1 Loss of C-V2X after CV has received an RCVW		
Verification Phase	II – Field Test Verification		
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed: - The administrator and algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. - The administrator shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the administrator and algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets. Results will be compared with 2.2.1h to contrast the impact of the road conditions.		
Pass / Fail	☒ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)		Date: 7/26/2024
Test Case No. and Title	3.1 Loss of C-V2X after CV has received an RCVW		
Verification Phase	II – Field Test Verification		
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Available is displayed while the CV is moving across the HRI [RCVW-007] [VBS-001] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010]	Y		

Test Case No. and Title	3.1 Loss of C-V2X after CV has received an RCVW		
Verification Phase	II – Field Test Verification		
[VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]			
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]		N	The averages are 9.55 messages per second.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS-004a] [RBS-004b]	Y		
A System Fault alert is displayed/announced on the DVI as the C-V2X signal is detected to be lost. [VBS-018e] [VBS-018d] [VBS-019] [RBS-005]	Y		See Flickering Fault
Verify in the logs that the System Fault was generated due to a loss of C-V2X communication [VBS-018e]	Y		Specific message: "Error: SPAT Data Not Received"
Notes			

Test Case No. and Title	3.2 Loss of C-V2X Prior to entering the MAP
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: To demonstrate system's capability to detection and response to a loss of communications due to a failure with the C-V2X radio.
Requirements Verified	RCVW-007, VBS-001, 004, 006, 007a, 007b, 008, 010, 014, 018e, 018d, 019, 025, RBS-005, 007c
Brief Description	The purpose of this test is to demonstrate how the RCVW system will behave when there is a loss of C-V2X prior to the CV entering the approach zone. The system will generate a graphic and audio message informing the driver of a System fault when it detects that it was expecting to receive a C-V2X signal from a known HRI as it is approaching to it but failed to do so.
Test Setup and Configuration	Prior to beginning the test, the Test director shall verify that the following conditions are planned for this test: - The C-V2X radio will be disabled prior to the start of the test - The initial approach speed of the CV shall be 35 mph - Pavement surface condition is dry. - This test will be performed using a passenger car (type A). - No driver response, the vehicle advances through the HRI. The following configuration states should be verified prior to test execution: - Ensure the VBS, including the DVI, are powered and operating normally (Power and GPS) - CV Inspector is running - Logging is activated
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed: The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case.

Test Case No. and Title	3.2 Loss of C-V2X Prior to entering the MAP		
Verification Phase	II – Field Test Verification		
	The algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets.		
Pass / Fail	☒ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)		Date: 7/26/2024
Test Case No. and Title		3.2 Loss of C-V2X Prior to entering the MAP	
Verification Phase		II – Field Test Verification	
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]		N	The averages are 9.55 messages per second.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS- 004a] [RBS-004b]	Y		
A System Fault alert is displayed/announced on the DVI as the VBS detects the absence of an expected C-V2X signal [RBS-005] [VBS-018e] [VBS-018d] [VBS-019]	Y		See Flickering Fault
Verify in the logs that the System Fault was generated due to a loss of C-V2X communication [VBS-018e]	Y		Specific message: "Error: SPAT Data Not Received"
Notes			

Test Case No. and Title	3.3 Loss of RTK Fix (GPS)
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: The purpose of this test is to demonstrate how the RCVW system will behave when there is a loss of RTK Fix from the GNSS communication on approach with the Preemption Signal active.
Requirements Verified	RCVW-007,008 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 007c, 008, 010, 011b, 011c, 012, 013, 014, 016, 017, 018b, 019, 020, 025 RBS-007c
Brief Description	As the CV enter the approach zone, with an active preemption signal, an approach inform will be displayed/annunciated. An RCVW will then be displayed/annunciated. As the driver begins to respond, there is a loss of RTK Fix in the VBS. The system will generate a graphic and audio message informing the driver of a System fault.
Test Setup and Configuration	Prior to beginning the test, the Test Administrator shall verify that the following conditions are planned for this test: - Train preemption signal detected at/prior to CV Approach Zone entry - The initial approach speed of the CV shall be 35 mph - The status of the RCVW system is “Normal” - This test will be performed using a CV passenger car (type A). - Driver response will be executed upon issuance of the RCVW, but prior to driver action resolving the RCVW, GPS signal is lost The following configuration states should be verified prior to test execution: - Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) - CV Inspector is running - Logging is activated
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above.

Test Case No. and Title	3.3 Loss of RTK Fix (GPS)		
Verification Phase	II – Field Test Verification		
	After the test, the following shall be performed: - The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. - the algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the administrator and algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets.		
Pass / Fail	☐ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)		Date: 7/26/2024
Test Case No. and Title	3.3 Loss of RTK Fix (GPS)		
Verification Phase	II – Field Test Verification		
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]		N	
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]		N	

Test Case No. and Title	3.3 Loss of RTK Fix (GPS)		
Verification Phase	II – Field Test Verification		
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]		N	
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]		N	The averages are 9.55 messages per second.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS-004a] [RBS-004b]	Y		
A System Fault alert is displayed/announced on the DVI as the GPS signal is detected to be lost. [VBS-18a] [VBS-018b] [VBS-019]	Y		See Flickering Fault
Verify in the logs that the System Fault was generated due to a loss of GPS signal [VBS-018b]	Y		Specific message: "Error: RTK Data Not Received"
Notes			

Test Case No. and Title	3.4 Loss of MAP
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: The purpose of this test is to demonstrate how the RCVW system will behave when there is a loss of MAP on approach with the Preemption Signal active.
Requirements Verified	RCVW-007,008 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 007c, 008, 010, 011b, 011c, 012, 013, 014, 016, 017, 018c, 018e, 019, 020, 025 RBS-007c
Brief Description	The purpose of this test is to demonstrate how the RCVW system will behave when there is a loss of MAP on approach with the Preemption Signal active. As the CV enter the approach zone, with an active preemption signal, an approach inform will be displayed/annunciated. An RCVW will then be displayed/annunciated. As the driver begins to respond, there is a loss of MAP from the RBS. The system will generate a graphic and audio message informing the driver of a System Fault.
Test Setup and Configuration	Prior to beginning the test, the Test director shall verify that the following conditions are planned for this test: - Train preemption signal detected at/prior to CV Approach Zone entry - The initial approach speed of the CV shall be 35 mph - The status of the RCVW system is “Normal” - This test will be performed using a CV passenger car (type A). - Driver response will be executed upon issuance of the RCVW, but prior to driver action resolving the RCVW, MAP data is loss The following configuration states should be verified prior to test execution: - Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) - Logging is activated
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed:

Test Case No. and Title	3.4 Loss of MAP		
Verification Phase	II – Field Test Verification		
	- the algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. - the algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the administrator and algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets.		
Pass / Fail	☒ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)		Date: 7/26/2024
Test Case No. and Title		3.4 Loss of MAP	
Verification Phase		II – Field Test Verification	
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]	Y		

Test Case No. and Title	3.4 Loss of MAP		
Verification Phase	II – Field Test Verification		
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]		N	The averages are 9.55 messages per second.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS-004a] [RBS-004b]	Y		
A System Fault alert is displayed/announced on the DVI as the loss of MAP occurs. [VBS-018c] [VBS-018e][VBS-019]	Y		See Flickering Fault
Verify in the logs that the System Fault was generated due to a loss of MAP [VBS-018C]	Y		Specific message: "Error: MAP Data Not Received"
Notes			

Test Case No. and Title	3.5 Loss of IEEE-1570 Heartbeat
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: The purpose of this test is to demonstrate how the RCVW system will behave when there is a loss of IEEE 1570 Heartbeat Signal on approach with the Preemption Signal active.
Requirements Verified	RCVW-007,008 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 007c, 008, 010, 011b, 011c, 012, 013, 014, 016, 017, 18e, 019, 020, 025 RBS-005a, 007c
Brief Description	As the CV enter the approach zone, with an active preemption signal, an approach inform will be displayed/annunciated. An RCVW will then be displayed/annunciated. As the driver begins to respond, there is a loss of the connection to the IEEE-1570 device resulting in the loss of the IEEE-1570 Heartbeat. The system will generate a graphic and audio message informing the driver of a Heartbeat fault.
Test Setup and Configuration	Prior to beginning the test, the Test Administrator shall verify that the following conditions are planned for this test: - Train preemption signal detected at/prior to CV Approach Zone entry - The initial approach speed of the CV shall be 35 mph - The status of the RCVW system is “Normal” - This test will be performed using a CV passenger car (type A). - Driver response will be executed upon issuance of the RCVW, but prior to driver action resolving the RCVW, IEEE Heartbeat signal is lost The following configuration states should be verified prior to test execution: - Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) - Logging is activated
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed: The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case.

Test Case No. and Title	3.5 Loss of IEEE-1570 Heartbeat		
Verification Phase	II – Field Test Verification		
	The algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets.		
Pass / Fail	☒ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)		Date: 7/26/2024
Test Case No. and Title	3.5 Loss of IEEE-1570 Heartbeat		
Verification Phase	II – Field Test Verification		
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]	Y		
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007]	Y		

Test Case No. and Title	3.5 Loss of IEEE-1570 Heartbeat		
Verification Phase	II – Field Test Verification		
[VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]			
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]		N	The averages are 9.55 messages per second.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS-004a] [RBS-004b]	Y		
A System Fault alert is displayed/announced on the DVI as the loss of IEEE 1570 Heartbeat signal occurs. [VBS-018e] [VBS-19] [RBS-16]	Y		See Flickering Fault
Verify in the logs that that System Fault was generated due to a loss of IEEE 1570 Heartbeat [RBS-10] [RBS-005a]	Y		Specific message: "Error: SPAT Data Not Received"
Notes			

Test Case No. and Title	4.1 Radio Range
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: The purpose of this test is to observe the C-V2X radio connectivity range.
Requirements Verified	N/A
Brief Description	The CV will be positioned at various locations and distances from the RSU to observe the radio connectivity range as the vehicle makes its way towards the RBS location.
Test Setup and Configuration	Prior to beginning the test, the Test Administrator shall verify that the following conditions are planned for this test: - RBS is on and broadcasting crossing information - VBS is on and OBU is ready to receive crossing information The following configuration states should be verified prior to test execution: - Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) - Logging is activated - Increase the distance of the “Distance to HRI” parameter in the VBS from the default 480 meters to 1 km.
Test Procedure/Script	The CV will approach the RBS from 3 separate locations. These locations will be within the 1 km radius covered in the “Distance to HRI” parameter. The VBS will expect to receive MAP and SPAT from the HRI since it will be located within the radius set. If a signal is not received, the VBI will issue a system fault. When the signal is received, the VBS will issue a System Ready message. The CV test observer will note the times when a System Ready and System Fault message were issued. The system logs will then be analyzed to extract latitude and /longitude locations. These locations will then be plotted on a map to calculate distances.

Test Case No. and Title	4.1 Radio Range		
Verification Phase	II – Field Test Verification		
	The start of the approach shall be a location where the CV is out of range of the radio and make its way towards the RBS. Location 1.- South bound on Route 142 Location 2.- North bound on Route 142 Location 3.- Battelle main access. See Appendix B for a Map of the proposed locations. These locations may change as the field test takes place.		
Pass / Fail	☒ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)		Date: 7/26/2024
Test Case No. and Title		4.1 Radio Range	
Verification Phase		II – Field Test Verification	
Expected Results [requirement]	Met?		Notes
	Y	N	
See Section Radio Range	N/A	N/A	
Notes			

Test Case No. and Title	4.2 GNSS Stability and Accuracy (Observation)
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: • Validate the stability of the GNSS System by confirming the following: ○ Confirm that a system available is displayed when the vehicle is positioned at the edge of the HRI and remains in the same state as the vehicle remains static in the location. ○ Confirm that a Clear HRI alert is displayed and remain when the vehicle is positioned within the HRI and remains in the same state as the vehicle remains static in the location. •
Requirements Verified	Specific requirements will not be validated by this test case. The test case will provide information related to the stability of the GNSS signal.
Brief Description	In this test case, the CV will approach the grade crossing, and stop at various locations prior to, within and beyond the HRI Hazard Zones (stop bars). This test case will validate the following system characteristics: • The stability and accuracy of the GPS System by cross checking the locations in the log where the different HMI state changes (alerts) display relative to the actual physical location of the entrance and exit of the HRI.
Test Setup and Configuration	Prior to beginning the test, the Test Administrator shall verify that the following conditions are planned for this test: • RBS is on and broadcasting crossing information • VBS is on and OBU is ready to receive crossing information The CV will be positioned in the following locations • Entire vehicle is within the Approach Zone, just outside of the HRI Hazard Zone

Test Case No. and Title	4.2 GNSS Stability and Accuracy (Observation)		
Verification Phase	II – Field Test Verification		
	• Front of vehicle inside the HRI Hazard Zone and rear of vehicle in the Approach Zone • Entire vehicle inside the HRI Hazard Zone • Front of the vehicle is beyond the HRI Hazard Zone and rear of the vehicle inside the HRI Hazard Zone • Entire vehicle is beyond (outside of) the HRI Hazard Zone The following configuration states should be verified prior to test execution: • Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) • Logging is activated		
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s). An external observer will assist the driver in stopping at each of the locations specified above. The CV observer will time out the duration that the vehicle should be stopped. At each location, the test observer will record the times any message was displayed on the DVI for later comparison with logs. After the test, the administrator and algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets.		
Pass / Fail	☒ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)		Date: 7/26/2024
Test Case No. and Title		4.2 GNSS Stability and Accuracy (Observation)	
Verification Phase		II – Field Verification	
Expected Results [requirement]	Met?		Notes
	Y	N	
When a part of the vehicle is within the HRI, an alert should be issued. This alert should be	Y		

Test Case No. and Title	4.2 GNSS Stability and Accuracy (Observation)		
Verification Phase	II – Field Test Verification		
extinguished when the vehicle moves and when the vehicle is outside the HRI location.			
Notes			

Regression Testing Results

Test Case No and Title	1.2 REGRESSION Type A CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: • Confirm that the System Ready graphic is displayed when the CV is outside the Approach Zone and system functioning • Confirm that the active preemption can be detected by the RBS by using a voltage-based connection and the IEEE-1570 interface. • Confirm that an Approach Inform Alert is displayed/annunciated while the CV is within the approach zone and approaching the HRI, there is an active Preemption signal and an RCVW is not predicted • Confirm that an RCVW is displayed/annunciated while the CV is within the Approach zone and approaching the HRI, there is an active Preemption Signal, and a RCVW is predicted • Confirm that an RCVW is no longer displayed/annunciated while the CV is within the Approach zone and approaching the HRI as a result of Driver Action (CV decelerates), there is an active Preemption Signal, and a RCVW is not/no longer predicted • Confirm that an Approach Inform Alert will be reissued while the CV is in the Approach Zone, preemption is active and RCVW is not predicted • Confirm that the Approach Inform Alert is extinguished after preemption deactivates. • Confirm that a System Active graphic is displayed while the CV is moving across the HRI (if applicable) • Confirm that a System Ready is displayed once the CV has departed the HRI (if applicable) • Analyze log files to confirm that warnings and alerts are accurate and correspond to displayed results
Requirements Verified	RCVW-007 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 008, 010, 011b, 011c, 012, 013, 014, 016, 017, 020, 021, 022, 023, 025 RBS-002b, 004, 004a, 004b, 007, 007a, 007b, 007c, 019, 022, 022a

Test Case No and Title	1.2 REGRESSION Type A CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
Brief Description	The primary purpose of this test is to verify that an RCVW is generated with the CV within the Approach zone and approaching the HRI, an active Preemption Signal, and a RCVW predicted for the specified conditions. Additionally, the test is to verify that the RCVW is no longer generated once the driver takes satisfactory action (CV decelerates) to permit safe stoppage prior to the HRI. This test case will verify that the RBS is capable of detecting the preemption signal by using a voltage-based connection and the IEEE-1570 interface. Three iterations using the IEEE-1570 device will be conducted following the same conditions described in this test case to collect multiple sets of data for later analysis.
Test Setup and Configuration	Prior to beginning the test, the Test director shall verify that the following conditions are planned for this test: • HRI status Active for the duration of this test • IEEE-1570 device are ready • The initial approach speed of the CV shall be 35 mph • Pavement surface condition is dry. • The status of the RCVW system is “Normal” • This test will be performed using a passenger car (type A). The following configuration states should be verified prior to test execution: • Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) • Ensure the RBS is powered and operating normally (MAP and SPaT are broadcast) • Logging is activated
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed: • The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case.

Test Case No and Title	1.2 REGRESSION Type A CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
	The algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case		
Pass/Fail	☒ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)		Date: 7/25/2024
Test Case No and Title	1.2 Type A CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verifications		
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
A System Available is displayed while the CV is moving across the HRI [RCVW-007] [VBS-001] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]	Y		
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-011d] [VBS-012a] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]	Y		

Test Case No and Title	1.2 REGRESSION Type A CV Approaches with Preemption Signal Detected at Approach Entry, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
The RCVW will extinguish after proper driver reaction and an Approach Inform Alert will be reissued while the CV is in the Approach Zone, preemption is active and no RCVW is not predicted [VBS-011b]	Y		
The RBS shall be capable of generating and broadcasting RTK corrections using the RTCM messaging protocol. [RBS-022] [RBS-022a]	Y		
Notes			

Test Case No. and Title	1.3 REGRESSION Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: • Confirm that the System Ready graphic is displayed when the CV is outside the Approach Zone and system functioning • Confirm that a System Available is displayed as the CV enters the approach zone and the preemption signal is not active • Confirm that an Approach Inform Alert is displayed/annunciated while the CV is within the approach zone and approaching the HRI, there is an active Preemption signal and an RCVW is not predicted • Confirm that an RCVW is displayed/annunciated while the CV is within the Approach zone and approaching the HRI, there is an active Preemption Signal, and a RCVW is predicted • Confirm that an RCVW is no longer displayed/annunciated while the CV is within the Approach zone and approaching the HRI as a result of Driver Action (CV decelerates), there is an active Preemption Signal, and a RCVW is not/no longer predicted • Confirm that an Approach Inform Alert will be reissued while the CV is in the Approach Zone, preemption is active and RCVW is not predicted • Confirm that the Approach Inform Alert is extinguished after preemption deactivates. • Confirm that a System Available is displayed while the CV is moving across the HRI (if applicable) • Confirm that a System Ready is displayed once the CV has departed the HRI (if applicable) • Analyze log files to confirm that warnings and alerts are accurate and correspond to displayed results
Requirements Verified	RCVW-007 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 008, 010, 011b, 011c, 012, 013, 014, 016, 017, 020, 021, 022, 023, 025 RBS-004, 004a, 004b, 007, 007a, 007b, 007c, 019, 022, 022a
Brief Description	The primary purpose of this test is to verify that an RCVW is generated with the CV within the Approach zone and approaching the HRI, an active Preemption Signal, and a RCVW predicted for the specified conditions. Additionally, the test is to verify that the RCVW is no longer generated once the driver takes satisfactory action to permit safe stoppage prior to the HRI.

Test Case No. and Title	1.3 REGRESSION Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions
Verification Phase	II – Field Test Verification
	This Test Case is similar to Test Case 1.2 except that the preemption detection becomes active when the CV is within the Approach Zone at a point $\frac{3}{4}$ the distance of the zone (based on initial approach entry speed), rather than at entry into the Approach Zone. 3 iterations will be conducted following the same conditions described in this test case to collect multiple sets of data for later analysis.
Test Setup and Configuration	Prior to beginning the test, the Test director shall verify that the following conditions are planned for this test: • HRI status Active when the CV reaches the 75% Approach Zone point • The initial approach speed of the CV shall be 35 mph • Pavement surface condition is dry. • The status of the RCVW system is “Normal” • This test will be performed using a passenger car (type A). The following configuration states should be verified prior to test execution: • Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) • Ensure the RBS is powered and operating normally (MAP and SPaT are broadcast) • Logging is activated
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 Error! Reference source not found.above. After the test, the following shall be performed: • Coordination between the driver and test director via radio will take place to coordinate when the CV approaches the $\frac{3}{4}$ distance so the preemption can be activated. • The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. • The algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case

Test Case No. and Title	1.3 REGRESSION Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
	After the test, the algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets. Results will be compared with test case 1.2 to contrast the impact of the distance from HRI when Preemption Signal is detected		
Pass / Fail	☒ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)		Date: 12/13/2024
Test Case No. and Title	1.3 Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
A System Available is displayed while the CV is moving across the HRI [RCVW-007] [VBS-001] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning. [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-012]	Y		

Test Case No. and Title	1.3 REGRESSION Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
[VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]			
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-011d] [VBS-012a] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]	Y		
The RCVW will extinguish after proper driver reaction and an Approach Inform Alert will be reissued while the CV is in the Approach Zone, preemption is active and no RCVW is not predicted [VBS-011b]	Y		
RBS broadcasts the HRI Geometry once per second and indicate the HRI ID [RBS-007] [RBS-007a] [RBS-007b]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]			Not verified in this test event
The HRI State message contains the HRI ID and the state of the preemption signal [RBS-004a] [RBS-004b]	Y		
The VBS shall be capable of receiving messages sent by the RBS within 50 ms.[VBS-022]			Not verified by this test event
The VBS shall be capable of processing received data within 85 ms.[VBS-023]			Not verified by this test event.

Test Case No. and Title	1.3 REGRESSION Type A CV Approaches with Preemption Signal Detected at the $\frac{3}{4}$ Approach Point, 35 mph, Dry Conditions		
Verification Phase	II – Field Test Verification		
The RBS shall be capable of generating and broadcasting RTK corrections using the RTCM messaging protocol. [RBS-022] [RBS-022a]	Y		
All log files shall be time stamped. [6]	Y		
Notes	As expected, having a delay on the detection of a train as the vehicle is approaching the HRI, requires a quicker action by the driver. However, as experienced before, the RCVW system provides the warning at the right time.		

Test Case No. and Title	2.1.REGRESSION CV Stopped in HRI Hazard Zone, Preemption not active/active
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: • Confirm that a System Ready is displayed when the CV is outside the Approach Zone and system functioning • Confirm that a system available is displayed when the CV is in the Approach Zone • Confirm that a Unique Warn alert to clear the HRI is displayed when the CV is stopped in the HRI and the preemption is active. • Confirm that a Unique alert to clear the HRI is displayed when the CV is stopped in the HRI and the preemption is not active. • Confirm that a System Ready is displayed when the CV leaves the HRI
Requirements Verified	RCVW-007 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 008, 010, 011b, 011c, 011d, 012a,012, 013, 014, 016, 017, 020, 021, 022, 023, 025, 026 RBS-004, 004a, 004b, 007, 007a, 007b, 007c, 019, 022, 022a
Brief Description	In this test case, the CV will approach the grade crossing, and then stop on the tracks (in the HRI Hazard Zone). A Unique Warn alert to clear the HRI shall be issued for the vehicle stopped in the HRI Hazard zone once the preemption is switched on. The warning will be removed once preemption is turned off, and replaced with a unique Alert to vacate the HRI. All messages are removed once the CV departs the HRI Hazard Zone. 3 iterations of each case will be conducted following the same conditions described in this test case to collect multiple sets of data for later analysis
Test Setup and Configuration	Prior to beginning the test, the Test director shall verify that the following conditions are planned for this test: • The CV approaches the HRI and stops on the tracks in the HRI Hazard Zone • With preemption off, an alert will be issued to vacate the HRI • With preemption on, a warning will be issued to vacate the HRI • The CV will depart the HRI after the alert and warning have been issued • The status of the RCVW system is “Normal” • This test will be performed using a passenger car (type A). The following configuration states should be verified prior to test execution:

Test Case No. and Title	2.1.REGRESSION CV Stopped in HRI Hazard Zone, Preemption not active/active		
Verification Phase	II – Field Test Verification		
	• Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) • Ensure the RBS is powered and operating normally (MAP and SPaT are broadcast) • Logging is activated		
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5Error! Reference source not found. above. After the test, the following shall be performed: • The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. • the algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets.		
Pass / Fail	☐ Pass (met all expected results) ☒ Fail (did not meet one or more expected results)		Date: 7/26/2024
Test Case No. and Title		2.1. CV Stopped in HRI Hazard Zone, Preemption not active/active	
Verification Phase		II – Field Test Verification	
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
A System Available is displayed while the CV is moving across the HRI [RCVW-007] [VBS-001] [VBS-006] [VBS-007a] [VBS-007b] [VBS-	Y		

Test Case No. and Title	2.1.REGRESSION CV Stopped in HRI Hazard Zone, Preemption not active/active		
Verification Phase	II – Field Test Verification		
008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]			
The VBS shall continually issue unique HRI Hazard informs to the vehicle operator while the HRI safety devices are not active and the vehicle is stopped within the HRI Hazard Zone [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-011d] [VBS-012a] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025] [VBS-012a]		N	The RCVW does provide an alert to driver urging to remove the vehicle from the HRI hazard zone. However, the requirements state that a different alert should be issued depending on the status of the HRI Safety devices. This did not occurred.
The VBS shall continually issue unique HRI Hazard warnings to the vehicle operator while the HRI safety devices are active and the vehicle is stopped within the HRI Hazard Zone. [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-011d] [VBS-012a] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025] [VBS-11d]		N	The RCVW does provide an alert to driver urging to remove the vehicle from the HRI hazard zone. However, the requirements state that a different alert should be issued depending on the status of the HRI Safety devices. This did not occurred.
Confirm that the CV will not receive an Alert Warning it of clearing the HRI as its moving through it	Y		
Confirm that the System Available message remains displayed while any portion of the vehicle occupies the HRI and the System Ready message displays only when all portions of the vehicle are clear of the HRI Approach Zone and HRI Hazard Zone [VBS-026]	Y		

Test Case No. and Title	2.1.REGRESSION CV Stopped in HRI Hazard Zone, Preemption not active/active		
Verification Phase	II – Field Test Verification		
RBS broadcasts the HRI Geometry once per second and indicate the HRI ID [RBS-007] [RBS-007a] [RBS-007b]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]			Not verified by this test event.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS-004a] [RBS-004b]	Y		
The VBS shall be capable of receiving messages sent by the RBS within 50 ms.[VBS-022]			Not verified by this test event.
The VBS shall be capable of processing received data within 85 ms.[VBS-023]	Y		
The RBS shall be capable of generating and broadcasting RTK corrections using the RTCM messaging protocol. [RBS-022] [RBS-022a]			Not verified by this test event.
All log files shall be time stamped. [RBS-019]	Y		
Notes	There was not a differentiation in the warning message issued when the HRI warning devices were activated or deactivated,		

Test Case No. and Title	3.1 REGRESSION Loss of C-V2X after CV has received an RCVW
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: To demonstrate system's capability to detection and response to a loss of communications due to a failure with the C-V2X radio.
Requirements Verified	RCVW-007, VBS-001, 002, 003a, 003b, 003c, 006, 007, 007a, 007b, 008, 010, 011b, 011c, 012, 013014, 016, 017,018e, 018d, 019, 020, 025 RBS-005, 007c
Brief Description	The purpose of this test is to demonstrate how the RCVW system will behave when there is a loss of C-V2X after the system has issued an RCVW. As the CV enter the approach zone, with an active preemption signal, an approach inform will be displayed/annunciated. An RCVW will then be displayed/annunciated. As the driver begins to respond, there is a loss of C-V2X signal. The system will generate a graphic and audio message informing the driver of a System fault.
Test Setup and Configuration	Prior to beginning the test, the Test Director shall verify that the following conditions are planned for this test: - Train preemption signal detected at/prior to CV Approach Zone entry - The initial approach speed of the CV shall be 35 mph - The status of the RCVW system is "Normal" - This test will be performed using a CV passenger car (type A). - Driver response will be executed upon issuance of the RCVW, but prior to driver action resolving the RCVW C-V2X communications are lost. The following configuration states should be verified prior to test execution: - Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) - Logging is activated

Test Case No. and Title	3.1 REGRESSION Loss of C-V2X after CV has received an RCVW		
Verification Phase	II – Field Test Verification		
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed: - The administrator and algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. - The administrator shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the administrator and algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets. Results will be compared with 2.2.1h to contrast the impact of the road conditions.		
Pass / Fail	☒ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)		Date: 12/12/2024
Test Case No. and Title		3.1 Loss of C-V2X after CV has received an RCVW	
Verification Phase		II – Field Test Verification	
Expected Results [requirement]	Met?		Notes
	Y	N	
A System Available is displayed while the CV is moving across the HRI [RCVW-007] [VBS-001] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010]	Y		

Test Case No. and Title	3.1 REGRESSION Loss of C-V2X after CV has received an RCVW		
Verification Phase	II – Field Test Verification		
[VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]			
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]			Not verified by this test event.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS-004a] [RBS-004b]	Y		
A System Fault alert is displayed/announced on the DVI as the C-V2X signal is detected to be lost. [VBS-018e] [VBS-018d] [VBS-019] [RBS-005]	Y		See Flickering Fault
Verify in the logs that the System Fault was generated due to a loss of C-V2X communication [VBS-018e]	Y		Specific message: "Error: SPAT Data Not Received"
Notes		The existing Software bug that results on the issuing and clearing fault messages is still present.	

Test Case No. and Title	3.3 REGRESSION Loss of RTK Fix (GPS)
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: The purpose of this test is to demonstrate how the RCVW system will behave when there is a loss of RTK Fix from the GNSS communication on approach with the Preemption Signal active.
Requirements Verified	RCVW-007,008 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 007c, 008, 010, 011b, 011c, 012, 013, 014, 016, 017, 018b, 019, 020, 025 RBS-007c
Brief Description	As the CV enter the approach zone, with an active preemption signal, an approach inform will be displayed/annunciated. An RCVW will then be displayed/annunciated. As the driver begins to respond, there is a loss of RTK Fix in the VBS. The system will generate a graphic and audio message informing the driver of a System fault.
Test Setup and Configuration	Prior to beginning the test, the Test Administrator shall verify that the following conditions are planned for this test: - Train preemption signal detected at/prior to CV Approach Zone entry - The initial approach speed of the CV shall be 35 mph - The status of the RCVW system is “Normal” - This test will be performed using a CV passenger car (type A). - Driver response will be executed upon issuance of the RCVW, but prior to driver action resolving the RCVW, GPS signal is lost The following configuration states should be verified prior to test execution: - Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) - CV Inspector is running - Logging is activated
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed:

Test Case No. and Title	3.3 REGRESSION Loss of RTK Fix (GPS)		
Verification Phase	II – Field Test Verification		
	- The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. - the algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the administrator and algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets.		
Pass / Fail	☒ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)	Date: 12/12/2024-12/13/2024	
Test Case No. and Title		3.3 Loss of RTK Fix (GPS)	
Verification Phase		II – Field Test Verification	
Expected Results [requirement]	Met?		Notes: GPS stability issues were experienced during the regression test event. RTK fix was periodically lost, which resulted in the issuing of the System Fault. The existing Software bug that results on the issuing and clearing fault messages is still present.
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-	Y		

Test Case No. and Title	3.3 REGRESSION Loss of RTK Fix (GPS)		
Verification Phase	II – Field Test Verification		
010] [VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]			
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]			Not verified by this test event.
The HRI State message contains the HRI ID and the state of the preemption signal [RBS-004a] [RBS-004b]			Not verified by this test event.
A System Fault alert is displayed/announced on the DVI as the GPS signal is detected to be lost. [VBS-18a] [VBS-018b] [VBS-019]	Y		See Flickering Fault
Verify in the logs that the System Fault was generated due to a loss of GPS signal [VBS-018b]	Y		Specific message: "Error: RTK Data Not Received"
Notes			

Test Case No. and Title	3.4 REGRESSION Loss of MAP
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: The purpose of this test is to demonstrate how the RCVW system will behave when there is a loss of MAP on approach with the Preemption Signal active.
Requirements Verified	RCVW-007,008 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 007c, 008, 010, 011b, 011c, 012, 013, 014, 016, 017, 018c, 018e, 019, 020, 025 RBS-007c
Brief Description	The purpose of this test is to demonstrate how the RCVW system will behave when there is a loss of MAP on approach with the Preemption Signal active. As the CV enter the approach zone, with an active preemption signal, an approach inform will be displayed/annunciated. An RCVW will then be displayed/annunciated. As the driver begins to respond, there is a loss of MAP from the RBS. The system will generate a graphic and audio message informing the driver of a System Fault.
Test Setup and Configuration	Prior to beginning the test, the Test director shall verify that the following conditions are planned for this test: - Train preemption signal detected at/prior to CV Approach Zone entry - The initial approach speed of the CV shall be 35 mph - The status of the RCVW system is “Normal” - This test will be performed using a CV passenger car (type A). - Driver response will be executed upon issuance of the RCVW, but prior to driver action resolving the RCVW, MAP data is loss The following configuration states should be verified prior to test execution: - Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) - Logging is activated
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed:

Test Case No. and Title	3.4 REGRESSION Loss of MAP		
Verification Phase	II – Field Test Verification		
	- the algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case. - the algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the administrator and algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets.		
Pass / Fail	☒ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)		Date: 12/12/2024
Test Case No. and Title		3.4 Loss of MAP	
Verification Phase		II – Field Test Verification	
Expected Results [requirement]	Met?		Notes: The existing Software bug that results on the issuing and clearing fault messages is still present.
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]	Y		

Test Case No. and Title	3.4 REGRESSION Loss of MAP		
Verification Phase	II – Field Test Verification		
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]	Y		
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]			Not verified in this test event
The HRI State message contains the HRI ID and the state of the preemption signal [RBS-004a] [RBS-004b]	Y		
A System Fault alert is displayed/announced on the DVI as the loss of MAP occurs. [VBS-018c] [VBS-018e][VBS-019]	Y		See Flickering Fault
Verify in the logs that the System Fault was generated due to a loss of MAP [VBS-018C]	Y		Specific message: "Error: MAP Data Not Received"
Notes			

Test Case No. and Title	3.5 REGRESSION Loss of IEEE-1570 Heartbeat
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: The purpose of this test is to demonstrate how the RCVW system will behave when there is a loss of IEEE 1570 Heartbeat Signal on approach with the Preemption Signal active.
Requirements Verified	RCVW-007,008 VBS-001, 002, 003a, 003b, 003c, 004, 006, 007, 007a, 007b, 007c, 008, 010, 011b, 011c, 012, 013, 014, 016, 017, 18e, 019, 020, 025 RBS-005a, 007c
Brief Description	As the CV enter the approach zone, with an active preemption signal, an approach inform will be displayed/annunciated. An RCVW will then be displayed/annunciated. As the driver begins to respond, there is a loss of the connection to the IEEE-1570 device resulting in the loss of the IEEE-1570 Heartbeat. The system will generate a graphic and audio message informing the driver of a Heartbeat fault.
Test Setup and Configuration	Prior to beginning the test, the Test Administrator shall verify that the following conditions are planned for this test: - Train preemption signal detected at/prior to CV Approach Zone entry - The initial approach speed of the CV shall be 35 mph - The status of the RCVW system is “Normal” - This test will be performed using a CV passenger car (type A). - Driver response will be executed upon issuance of the RCVW, but prior to driver action resolving the RCVW, IEEE Heartbeat signal is lost The following configuration states should be verified prior to test execution: - Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) - Logging is activated
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s) per Table 5 above. After the test, the following shall be performed: The algorithm observer shall ensure the system did not enter an error state, system is active and ready for the next test case.

Test Case No. and Title	3.5 REGRESSION Loss of IEEE-1570 Heartbeat		
Verification Phase	II – Field Test Verification		
	The algorithm observer shall confirm that data was properly recorded by all test personnel and indicate to proceed to next iteration/test case After the test, the algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets.		
Pass / Fail	☒ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)		Date: 12/13/2024
Test Case No. and Title		3.5 Loss of IEEE-1570 Heartbeat	
Verification Phase		II – Field Test Verification	
Expected Results [requirement]	Met?		Notes: A new bug has been identified Once the IEEE Heartbeat was lost, the test team was not able to recover. The existing Software bug that results on the issuing and clearing fault messages is still present.
	Y	N	
A System Ready is displayed while the CV is outside of the approach zone [RCVW-007] [VBS-001] [VBS-004] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-014] [VBS-025] [RBS-007c]	Y		
An Approach Inform Alert is displayed/announced while the CV is within the approach zone and approaching the HRI with an active Preemption Signal while vehicle speed and distance do not warrant an approach warning [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-004] [VBS-006] [VBS-007] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-012] [VBS-013] [VBS-014] [VBS-016] [VBS-017] [VBS-025]	Y		
An RCVW is displayed/annunciated while the CV is within the approach zone and approaching the HRI with an active	Y		

Test Case No. and Title	3.5 REGRESSION Loss of IEEE-1570 Heartbeat		
Verification Phase	II – Field Test Verification		
Preemption Signal throughout the approach before driver action is taken [RCVW-007] [VBS-001] [VBS-002] [VBS-003a] [VBS-003b] [VBS-003c] [VBS-006] [VBS-007a] [VBS-007b] [VBS-008] [VBS-010] [VBS-011b] [VBS-011c] [VBS-014] [VBS-016] [VBS-017] [VBS-020] [VBS-025]			
RBS broadcasts the HRI Activation Status message (SPaT) 10 times per second [RBS-004]			Not verified in this test event
The HRI State message contains the HRI ID and the state of the preemption signal [RBS-004a] [RBS-004b]	Y		
A System Fault alert is displayed/announced on the DVI as the loss of IEEE 1570 Heartbeat signal occurs. [VBS-018e] [VBS-19] [RBS-16]	Y		See Flickering Fault
Verify in the logs that that System Fault was generated due to a loss of IEEE 1570 Heartbeat [RBS-10] [RBS-005a]	Y		Specific message: "Error: SPAT Data Not Received"
Notes			

Test Case No. and Title	4.2 REGRESSION - GNSS Stability and Accuracy (Observation)
Verification Phase	II – Field Test Verification
Test Objectives	Primary Objectives: • Validate the stability of the GNSS System by confirming the following: ○ Confirm that a system available is displayed when the vehicle is positioned at the edge of the HRI and remains in the same state as the vehicle remains static in the location. ○ Confirm that a Clear HRI alert is displayed and remain when the vehicle is positioned within the HRI and remains in the same state as the vehicle remains static in the location. •
Requirements Verified	Specific requirements will not be validated by this test case. The test case will provide information related to the stability of the GNSS signal.
Brief Description	In this test case, the CV will approach the grade crossing, and stop at various locations prior to, within and beyond the HRI Hazard Zones (stop bars). This test case will validate the following system characteristics: • The stability and accuracy of the GPS System by cross checking the locations in the log where the different HMI state changes (alerts) display relative to the actual physical location of the entrance and exit of the HRI.
Test Setup and Configuration	Prior to beginning the test, the Test Administrator shall verify that the following conditions are planned for this test: • RBS is on and broadcasting crossing information • VBS is on and OBU is ready to receive crossing information The CV will be positioned in the following locations • Entire vehicle is within the Approach Zone, just outside of the HRI Hazard Zone

Test Case No. and Title	4.2 REGRESSION - GNSS Stability and Accuracy (Observation)		
Verification Phase	II – Field Test Verification		
	• Front of vehicle inside the HRI Hazard Zone and rear of vehicle in the Approach Zone • Entire vehicle inside the HRI Hazard Zone • Front of the vehicle is beyond the HRI Hazard Zone and rear of the vehicle inside the HRI Hazard Zone • Entire vehicle is beyond (outside of) the HRI Hazard Zone The following configuration states should be verified prior to test execution: • Ensure the VBS, including the DVI, are powered and operating normally (Power, GPS, and C-V2X) • Logging is activated		
Test Procedure/Script	Prior to beginning the test, the personnel will be located at their assigned position(s). An external observer will assist the driver in stopping at each of the locations specified above. The CV observer will time out the duration that the vehicle should be stopped. At each location, the test observer will record the times any message was displayed on the DVI for later comparison with logs. After the test, the administrator and algorithm observer shall access and review logged files to confirm they are accurate and correspond to observed results as recorded on data sheets.		
Pass / Fail	☒ Pass (met all expected results) ☐ Fail (did not meet one or more expected results)		Date: 12/13/2024
Test Case No. and Title		4.2 GNSS Stability and Accuracy (Observation)	
Verification Phase		II – Field Verification	
Expected Results [requirement]	Met?		Notes: When the system is able to achieve an RTK fix, these test all passed. See GNSS Stability .
	Y	N	
When a part of the vehicle is within the HRI, an alert should be issued. This alert should be	Y		

Test Case No. and Title	4.2 REGRESSION - GNSS Stability and Accuracy (Observation)		
Verification Phase	II – Field Test Verification		
extinguished when the vehicle moves and when the vehicle is outside the HRI location.			
Notes			

Appendix.

RCVW Requirements Traceability Matrix

Table A-1 below identifies the FT&E test case(s) in which each requirement from the RCVW System Requirements Specification document [2] is verified. Scenario-based tests for FT&E are a subset of those conducted in BT&E, but revised to reflect how field testing is conducted.

RCVW Rqmt No.	System/Subsystem	Requirement	Phase III RCVW FT&E Test Cases	Met?
			Scenario-Based	
RCVW-001	RCVW System	The system shall include a VBS component, a RBS component and a CBS component.	All scenarios	Y
RCVW-003	RCVW System	The system shall be modular and sufficiently extensible to address all design objectives defined in this SRS.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RCVW-005	RCVW System	The only point(s) of connection between the RCVW system and the HRI Controller shall be the preemption signal itself or via a third-party device that provides an IEEE 1570-compliant serial interface.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RCVW-007	RCVW System	The VBS On-board Unit (OBU) and RBS Roadside Unit (RSU) shall communicate in compliance with the suite of IEEE-1609 standards/protocols and SAE J2735_202211 Standards.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.2, 3.3, 3.4, 3.5	Y
RCVW-008	RCVW System	All "over-the-road" licensed vehicles (i.e., vehicles of all vehicle classes) are included.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RCVW-011	RCVW System	The system shall be compliant with CV Personally Identifiable Information (PII) standards and guidelines.	Verified in Phase II of the RCVW project.	N/A
RCVW-012	RCVW System	The RBS and the CBS shall communicate in compliance with SAE J2735_202211, IEEE 1609 over a TCP/IP based BCN.	All scenarios	Y
RCVW-013	RCVW System	All communications between the RBS and the VBS/CBS shall be secured through the use of authenticated certificates issued through the Security Credential Management System (SCMS).	Requirement moved to Production	N/A
RCVW-014	RCVW System	The start of an HRI Approach lane for the purposes of RCVW shall equate to a vehicle driver's visual range of a Grade Crossing as defined by the Guidelines for Advance Placement of Warning Signs in Table 2C-4 of the 2009 Manual on Uniform Traffic Control Devices (MUTCD), Revision 2, June 13, 2012.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A

Appendix A. RCVW Requirements Traceability Matrix

RCVW Rqmt No.	System/Subsystem	Requirement	Phase III RCVW FT&E Test Cases	Met?
			Scenario-Based	
VBS-001	Vehicle Based Subsystem (VBS)	The VBS shall have the capability to produce alerts suitable for all licensed drivers.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.2, 3.3, 3.4, 3.5	Y
VBS-002	Vehicle Based Subsystem (VBS)	The VBS shall have a human-machine interface (HMI) that is configurable to be audible, visual, both, or neither by the driver.	1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.3, 3.4, 3.5	Y
VBS-003a	Vehicle Based Subsystem (VBS)	The VBS will present alerts that conform to Campbell et al. (2004) In-Vehicle Display Icons and Other Information Elements, Volume 1: Guidelines and Campbell et al. (2016) Human Factors Design Guidance for Driver-Vehicle Interfaces.	1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.3, 3.4, 3.5	Y
VBS-003b	Vehicle Based Subsystem (VBS)	The system shall provide two-stage alert messaging consisting of an inform alert, and, if applicable, a warning.	1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.3, 3.4, 3.5	Y
VBS-003c	Vehicle Based Subsystem (VBS)	The inform and warning alerts shall be multimodal (visual and auditory) in nature.	1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.3, 3.4, 3.5	Y
VBS-004	Vehicle Based Subsystem (VBS)	The VBS shall produce alerts that can be implemented in all vehicle classes and types equipped with appropriate CV technologies.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.2, 3.3, 3.4, 3.5	Y
VBS-005	Vehicle Based Subsystem (VBS)	The VBS shall receive and process RTK corrections using the RTCM messaging protocol broadcasted from the RBS to achieve a R95 probability of horizontal position accuracy of less than or equal to 1.5 meters.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
VBS-006	Vehicle Based Subsystem (VBS)	The VBS shall process HRI geometry data to determine the geospatial location and the ingress and egress lanes for the HRI.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.2, 3.3, 3.4, 3.5	Y
VBS-007	Vehicle Based Subsystem (VBS)	The VBS shall only produce alerts when the vehicle is in range and approaching a known RCVW RBS equipped HRI.	1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.3, 3.4, 3.5	Y
VBS-007a	Vehicle Based Subsystem (VBS)	A vehicle is considered in range when its current position (accounting for all uncertainties) is within the boundaries of a configurable radius away from a known RCVW RBS equipped HRI.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 3.1, 3.2, 3.3, 3.4, 3.5	Y
VBS-007b	Vehicle Based Subsystem (VBS)	The configurable radius in VBS-007a shall not be less than 0.5 km.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 3.1, 3.2, 3.3, 3.4, 3.5	Y
VBS-008	Vehicle Based Subsystem (VBS)	The system shall utilize a driver-vehicle interface (DVI) and OEM displays through standard physical and electrical outputs.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 2.2, 3.1, 3.2, 3.3, 3.4, 3.5	Y
VBS-009a	Vehicle Based Subsystem (VBS)	The VBS shall not interfere with any of the onboard safety systems, especially automotive industry automated safety systems	Verified per manufacturer certifications.	N/A
VBS - 009b	Vehicle Based Subsystem (VBS)	The VBS shall not interfere with any existing infrastructure subsystems (traffic control and HRI warning systems)	Verified per manufacturer certifications.	N/A

Appendix A. RCVW Requirements Traceability Matrix

RCVW Rqmt No.	System/Subsystem	Requirement	Phase III RCVW FT&E Test Cases	Met?
			Scenario-Based	
VBS-010	Vehicle Based Subsystem (VBS)	The VBS shall determine if the vehicle is within the HRI Hazard Zone and/or the HRI Approach Zone.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 2.2, 3.1, 3.2, 3.3, 3.4, 3.5	Y
VBS-011b	Vehicle Based Subsystem (VBS)	The VBS shall continually issue unique HRI Approach warnings to the vehicle operator while the HRI safety devices are active and the vehicle is within the HRI Approach Zone but not decelerating sufficiently to stop safely before the HRI using non-emergency braking.	1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.3, 3.4, 3.5	Y
VBS-011c	Vehicle Based Subsystem (VBS)	All VBS <i>warnings</i> shall use a dynamic visual icon(s) and intrusive auditory alert(s).	1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.3, 3.4, 3.5	Y
VBS-011d	Vehicle Based Subsystem (VBS)	The VBS shall continually issue unique HRI Hazard warnings to the vehicle operator while the HRI safety devices are active and the vehicle is stopped within the HRI Hazard Zone.	2.1	Y
VBS-012	Vehicle Based Subsystem (VBS)	The VBS shall continually issue unique HRI Approach informational alert messages to the vehicle operator while the HRI safety devices are active and the vehicle is within the HRI Approach Zone.	1.2, 1.3, 1.4, 1.5, 1.6, 3.1, 3.3, 3.4, 3.5	Y
VBS-012a	Vehicle Based Subsystem (VBS)	The VBS shall continually issue unique HRI Hazard informs to the vehicle operator while the HRI safety devices are not active and the vehicle is stopped within the HRI Hazard Zone.	2.1	Y
VBS-013	Vehicle Based Subsystem (VBS)	All VBS inform alerts shall use static visual icons and audible alert(s).	1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.3, 3.4, 3.5	Y
VBS-014	Vehicle Based Subsystem (VBS)	The VBS shall process the HRI activation status message in the context of its position within the HRI Hazard Zone or HRI Approach Zone, and utilizing the vehicle dynamics data, shall determine if an alert or warning is warranted.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.2, 3.3, 3.4, 3.5	Y
VBS-015	Vehicle Based Subsystem (VBS)	Warnings and alerts shall be generated with regard to: 85th percentile driver response time, vehicle characteristics (i.e., vehicle class and antenna offset), and vehicle dynamics	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
VBS-016	Vehicle Based Subsystem (VBS)	The VBS shall only issue inform alerts and warnings when the positional information (while accounting for all uncertainties) places the vehicle inside the HRI Hazard Zone or the HRI Approach Zone, based on the present HRI geometry.	1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.2, 3.3, 3.4, 3.5	Y
VBS-017	Vehicle Based Subsystem (VBS)	Once issued, the graphical component of an inform alert will persist while the vehicle is within the approach zone, except when superseded by a warning or fault alert, or when the HRI becomes inactive.	1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.2, 3.3, 3.4, 3.5	Y

Appendix A. RCVW Requirements Traceability Matrix

RCVW Rqmt No.	System/Subsystem	Requirement	Phase III RCVW FT&E Test Cases	Met?
			Scenario-Based	
VBS-018	Vehicle Based Subsystem (VBS)	The VBS shall contain a database of geospatial locations and IDs for known RCVW RBS equipped HRIs.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
VBS-018a	Vehicle Based Subsystem (VBS)	The VBS shall continually issue Position Prevalence fault alerts to the vehicle operator when the vehicle is in range of an HRI but the VBS positional information is being received at a lower average rate than what the Positional Prevalence fault parameter indicates.	3.3	Y
VBS-018b	Vehicle Based Subsystem (VBS)	The VBS shall continually issue Position Resolution fault alerts to the vehicle operator when the vehicle is in range of an HRI but the VBS positional resolution remains inferior to floating point or fixed integer RTK.	3.3	Y
VBS-018c	Vehicle Based Subsystem (VBS)	The VBS shall continually issue HRI Geometry Prevalence fault alerts to the vehicle operator when the vehicle is in range of an HRI but the HRI Geometry information is being received at a consistent rate under once per second.	3.4	Y
VBS-018d	Vehicle Based Subsystem (VBS)	The VBS shall continually issue HRI Activation Status Prevalence faults to the vehicle operator when the vehicle is in range of an HRI but the HRI Activation Status is being received at a lower average rate than what the HRI Activation Status Prevalence fault parameter indicates.	3.1, 3.2	Y
VBS-018e	Vehicle Based Subsystem (VBS)	The VBS shall continually issue Communication fault alerts to the vehicle operator when the vehicle is in range of an HRI and HRI Geometry messages and/or HRI Activation Status messages are not being received	3.1, 3.2, 3.4	Y
VBS-018f	Vehicle Based Subsystem (VBS)	The number of messages per second received used to calculate the issuing of a Position Prevalence fault shall be a user configurable parameter.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
VBS-018g	Vehicle Based Subsystem (VBS)	The number of messages per second received used to calculate the issuing a HRI Activation Status Prevalence fault shall be a user configurable parameter.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
VBS-019	Vehicle Based Subsystem (VBS)	Fault alerts shall supersede all other annunciations.	3.1, 3.2, 3.3, 3.4, 3.5	Y
VBS-020	Vehicle Based Subsystem (VBS)	Warnings shall supersede inform alerts.	1.2, 1.3, 1.4, 1.5, 1.6, 2.1,	Y
VBS-022	Vehicle Based Subsystem (VBS)	The VBS shall be capable of receiving messages sent by the RBS within 50 ms.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1,	N
VBS-023	Vehicle Based Subsystem (VBS)	The VBS shall be capable of processing received data within 85 ms.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1	Y

RCVW Rqmt No.	System/Subsystem	Requirement	Phase III RCVW FT&E Test Cases	Met?
			Scenario-Based	
VBS-024	Vehicle Based Subsystem (VBS)	The VBS shall know the position of the GNSS antenna relative to the front of the vehicle and the rear of the vehicle.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
VBS-025	Vehicle Based Subsystem (VBS)	The VBS shall be considered within the HRI Approach Zone whenever the current positional information while accounting for all uncertainties falls within the boundaries of an ingress lane, based on the current HRI Geometry.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.2, 3.3, 3.4, 3.5	Y
VBS-026	Vehicle Based Subsystem (VBS)	The VBS shall be considered within the HRI Hazard Zone whenever the current positional information while accounting for all uncertainties falls within the boundaries between all diametrically opposite lanes that also contains the geospatial location of the HRI, based on the current HRI Geometry.	2.1	Y
VBS-027	Vehicle Based Subsystem (VBS)	The RCVW VBS shall be designed to account for operation in all weather, road surface and visibility conditions.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RBS-001	Roadside-Based Subsystem (RBS)	The RBS shall interoperate with current infrastructure safety systems (e.g. traffic control and train detection devices) in accordance with NEMA TS 2-2021 v03.08.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RBS-002a	Roadside-Based Subsystem (RBS)	The RBS shall operate using 60 Hz 115VAC power as the primary power source.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RBS-002b	Roadside-Based Subsystem (RBS)	The RBS shall be capable of determining HRI activation status using the DC voltage-based preemption signal from the HRI Controller and, indirectly, from an IEEE 1570-compliant serial device that has been interfaced to the HRI Controller, when present.	1.2,	Y
RBS-003	Roadside-Based Subsystem (RBS)	All roadside-based subsystem messages shall contain a header segment that is compliant with the V2X Hub Reference Implementation platform.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RBS-003a	Roadside-Based Subsystem (RBS)	All RBS messages shall contain a payload segment containing all required information as an encoded sequence of bytes.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RBS-004	Roadside-Based Subsystem (RBS)	The RBS shall broadcast the HRI Activation Status message to both the VBS and CBS 10 times per second during operation.	All scenarios with the exception of 4.1 and 4.2	N
RBS-004a	Roadside-Based Subsystem (RBS)	The HRI State message shall uniquely identify the grade crossing it pertains to	All scenarios with the exception of 4.1 and 4.2	Y
RBS-004b	Roadside-Based Subsystem (RBS)	The HRI State message shall indicate the present state of the preemption signal.	All scenarios with the exception of 4.1 and 4.2	Y
RBS-005	Roadside-Based Subsystem (RBS)	The HRI State message shall indicate "active" whenever the detection system is off-line or disconnected.	3.1, 3.2	Y

RCVW Rqmt No.	System/Subsystem	Requirement	Phase III RCVW FT&E Test Cases	Met?
			Scenario-Based	
RBS-005a	Roadside-Based Subsystem (RBS)	When the RBS is connected to an IEEE-1570 compatible device and the Heartbeat signal is lost, the RBS shall cease sending HRI State Messages.	3.5	Y
RBS-007	Roadside-Based Subsystem (RBS)	The RBS shall broadcast the HRI Geometry to both the VBS and CBS once per second during operation.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1,	Y
RBS-007a	Roadside-Based Subsystem (RBS)	The HRI Geometry message shall uniquely identify the grade crossing it pertains to.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1	Y
RBS-007b	Roadside-Based Subsystem (RBS)	The HRI Geometry message shall provide the geospatial location of the HRI.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1	Y
RBS-007c	Roadside-Based Subsystem (RBS)	The HRI Geometry message shall provide the geospatial location and geometry of all lanes approaching to (ingress) and leading away from (egress) the HRI.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 3.1, 3.2, 3.3, 3.4, 3.5	Y
RBS-010	Roadside-Based Subsystem (RBS)	The RBS shall execute periodic BIST to determine the functional state of critical components.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RBS-011	Roadside-Based Subsystem (RBS)	The RBS shall employ methods to prevent unauthorized physical and cyber access.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RBS-012	Roadside-Based Subsystem (RBS)	The V2I communication shall implement security as defined by IEEE 1609 Standards for Wireless Access in the Vehicular Environment (WAVE). For clarity, a unique security solution will not be developed for this project, but the available security solution provided by U.S. DOT for V2I communications will be exercised.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RBS-013	Roadside-Based Subsystem (RBS)	Secure-communication protocols shall not adversely impact the performance of the safety application with respect to the ability to provide alerts in a timely manner.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RBS-016	Roadside-Based Subsystem (RBS)	The RBS shall identify and log system failures to the extent that it is practicable.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RBS-017	Roadside-Based Subsystem (RBS)	The RBS shall broadcast system diagnostics data to the CBS at least once per second during operation.		N/A
RBS-017a	Roadside-Based Subsystem (RBS)	The RBS Diagnostic message shall uniquely identify the HRI it pertains to.		N/A
RBS-17b	Roadside-Based Subsystem (RBS)	The RBS Diagnostic message shall report the current broadcast rate of the HRI Geometry messages.		N/A
RBS-017c	Roadside-Based Subsystem (RBS)	The RBS Diagnostic message shall report the current broadcast rate of the HRI State messages.		N/A

RCVW Rqmt No.	System/Subsystem	Requirement	Phase III RCVW FT&E Test Cases	Met?
			Scenario-Based	
RBS-017d	Roadside-Based Subsystem (RBS)	The RBS Diagnostic message shall report the current broadcast rate of the RTCM messages.		N/A
RBS-017e	Roadside-Based Subsystem (RBS)	The RBS Diagnostic message shall report the current operational state of all RBS components and overall performance.		N/A
RBS-018	Roadside-Based Subsystem (RBS)	The RBS shall incorporate self-recovering routines to recover from a major system failure associated with firmware/software systems.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RBS-019	Roadside-Based Subsystem (RBS)	Application and system logs will be timestamped using an internally consistent mechanism (e.g., GPS or UTC time) and maintained at least until reported through the RBS Diagnostic message.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1, A.1, A.2, A.3, A.4	Y
RBS-020	Roadside-Based Subsystem (RBS)	The RBS shall not interfere with any HRI infrastructure subsystems.	Verified per manufacturer certifications.	N/A
RBS-021	Roadside-Based Subsystem (RBS)	The RBS shall be capable of receiving HRI message packets across an IEEE 1570-compliant serial communication interface in less than 250 ms conforming to the IEEE-1570 standard.	Verified in Phase II of the RCVW project. N/A for this test event.	N/A
RBS-022	Roadside-Based Subsystem (RBS)	The RBS shall broadcast Radio Technical Commission for Maritime Services (RTCM) correction messages to the VBS at the same rate it receives them, but not more than once per second.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1	N
RBS-022a	Roadside-Based Subsystem (RBS)	The RBS shall be able to produce RTCM messages directly through the use of a local base station.	1.1, 1.2, 1.3, 1.4, 1.5, 1.6,	N
RBS-022b	Roadside-Based Subsystem (RBS)	The RBS shall be able to receive, and relay RTCM messages upon receiving them from the CBS.		N/A

Abbreviations and Acronyms

ACRONYMS	EXPLANATION
AASHTO	American Association of State Highway and Transportation
BT&E	Bench Test and Evaluation
C-V2X	Cellular Vehicle-to-Everything
CBS	Cloud Based Subsystem
CPU	Central Processing Unit
CP	Computing Platform
ConOps	Concept of Operations
CV	Connected Vehicle
DSRC	Dedicated Short-Range Communication
DVI	Driver-Vehicle Interface
FCC	Federal Communications Commission
FMVSS	Federal Motor Vehicle Safety Standards
FRA	Federal Railroad Administration
FT&E	Field Test & Evaluation
GNSS	Global Navigation Satellite Systems
GPS	Global Positioning System
HDMI	High-Definition Multimedia Interface
HRI	Highway-Rail Intersection
HDOP	Horizontal Dilution of Precision
IEEE	Institute of Electrical and Electronics Engineers
KM	Kilometer
MAP	MAP Data, Roadway Geometry and Attribute Data (aka GID)
MUTCD	Manual on Uniform Traffic Control Devices
OBU	On-Board Unit (DSRC radio in VBS)
PCAP	Paquet Capture
POE	Power Over Ethernet
PVT	Position-Velocity-Time
RBS	Roadside-Based Subsystem
RTCM	Radio Technical Commission for Maritime Services
RCVW	Rail Crossing Violation Warning

ACRONYMS**EXPLANATION**

RTK	Real-Time Kinematic
RBS	Roadside-Based Subsystem (an RCVW subsystem)
RSU	Roadside Unit (C-V2X radio in RBS)
SPaT	Signal Phase and Timing
TMX	Transportation Message Exchange
UTC	Universal Time Coordinated
VBS	Vehicle-Based Subsystem (an RCVW subsystem)
V2I	Vehicle-to-Infrastructure